Journal of Cleaner Production 235 (2019) 1276–1294

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Review

Analytic network process: Academic insights and perspectives analysis

Yong Chen [a], Qiuxia Jin [a], Hui Fang [b, c], Hui Lei [a], Jiarui Hu [a], Yanqi Wu [b, c], Jie Chen [b, c], Cheng Wang [a, *], Yuehua Wan [b, c, **]

[a] Institute of Industrial Engineering, Zhejiang University of Technology, Hangzhou, 310014, PR China
[b] Institute of Information Resources, Zhejiang University of Technology, Hangzhou, 310014, PR China
[c] Library, Zhejiang University of Technology, Hangzhou, 310014, PR China

ARTICLE INFO

Article history:
Received 7 January 2019
Received in revised form
29 May 2019
Accepted 2 July 2019
Available online 8 July 2019

Handling editor: Sandra Caeiro

Keywords:
Analytic Network Process
Web of science
Bibliometrics
Hot topics
Sustainability
Environmental management
Supply chain management

ABSTRACT

Diversity multi-criteria decision-making methods have been developed to address different complex decision-making problems, and the analytic network process has been found to be one of the most effective techniques. There is an increase in the quality and quantity of publications related to the analytic network process. This detailed overview can provide the research status and development characteristics of analytic network process research and will be useful to researchers for future research directions. To achieve these goals, bibliometric techniques were used. In addition, past and present hotspots of analytic network process research were concluded, and future research trends were determined. The bibliometric analysis was carried out from various aspects including countries and regions, institutions, journals, authors, research areas, articles and author keywords based on data harvested from the Web of Science database. There were 1485 analytic network process-related publications retrieved from the Web of Science. The results show that *Expert Systems with Applications* was the most productive journal publishing articles in analytic network process research (118); its number of publications has decreased dramatically since 2013, while *Journal of Cleaner Production* has shown an upward trend in recent years and ranks second with 47 publications. The most collaborative country is the United States. Taiwan takes a leading position in analytic network process research with 436 publications (29.36%), and National Chiao Tung University, which is located in Taiwan, produced the most articles and has gained the highest *h*-index (28). The major hot topics that employ analytic network process are sustainability, environmental management and supply chain management. These topics may continue to attract more attention in the future.

© 2019 Elsevier Ltd. All rights reserved.

Contents

1. Introduction 1277
   1.1. Data collection and analysis methods 1278
   1.2. Paper collection 1278
   1.3. Data collection 1279
   1.4. Analysis methods 1279
2. Results and discussions 1279
   2.1. Contribution of leading research areas 1280
   2.2. Leading journals that have contributed to the ANP 1281
   2.3. Analysis of author keywords 1283

* Corresponding author.
** Corresponding author. Institute of Information Resources, Zhejiang University of Technology, Hangzhou, 310014, PR China
E-mail addresses: cwang@zjut.edu.cn (C. Wang), wanyuehua@zjut.edu.cn (Y. Wan).

https://doi.org/10.1016/j.jclepro.2019.07.016
0959-6526/© 2019 Elsevier Ltd. All rights reserved.

2.4. Global contribution and leading countries/regions . . . 1285
2.5. Contribution of leading institutions . . . 1285
2.6. Leading authors who contributed to the ANP . . . 1285
2.7. Analysis of most cited published documents . . . 1288
2.8. Analysis of hot papers and hot topics for each year . . . 1289
3. Implications . . . 1290
3.1. A research areas perspective . . . 1291
3.2. A cooperation perspective . . . 1291
3.3. A future application trends perspective . . . 1291
4. Conclusions and limitations . . . 1291
Acknowledgments . . . 1292
References . . . 1292

1. Introduction

Decision making is part of our daily lives, and most decision-making problems are multi-criteria decision-making (MCDM) problems that involve criteria and alternatives to choose from; the importance of the criteria are usually different, and the alternatives vary according to the decision-makers' preference for them on each criterion (Saaty, 2004a). Saaty and Ergu (2015) provided 16 criteria to help researchers evaluate diverse MCDM methods and choose the most efficient and worthwhile method to make different decisions.

The analytic network process is one of the MCDM methods was proposed by Saaty in 1996 (Saaty, 1996). Thomas L. Saaty was a Distinguished University Professor at the University of Pittsburgh and is also the inventor of the analytic hierarchy process (AHP) (Saaty, 1980). He was awarded the Institute for Operation Research and the Management Science (INFORMS) Impact Prize for his development of the AHP in 2008; and in 2011, the paper entitled "The Possibility of Group Welfare Functions" (Saaty and Vargas, 2005), which was co-authored with Luis G. Vargas, received the Hebert Simon Award for Outstanding Contribution in Information Technology and Decision Making. Saaty has made many contributions in the ANP field: he implemented the ANP for forecasting the probability of financial crises (Niemira and Saaty, 2004); three approaches were introduced to address dynamic decision-making problems (Saaty, 2007); and he also introduced a method to derive relative scales for intangible factors arranged in the AHP or ANP structure (Saaty, 2008). His great contributions to decision science provide effective tools and methods for solving decision problems, provide broader ideas and lay a solid theoretical foundation for the proposal of other creative decision-making methods. A special issue about AHP/ANP theory and its application was published in 2016 to celebrate Professor Thomas L. Saaty's 90th birthday (Kou and Ergu, 2016), which shows Professor Thomas L. Saaty's outstanding contribution and respectable status in the realm of the AHP/ANP and MCDM research. We are deeply saddened by his passing and would like to pay tribute to him and his great work through this paper.

The ANP has been widely applied in various areas, such as supply chain management (Kannan, 2018; Büyüközkan and Çifçi, 2011), waste management (Coban et al., 2018; Liu et al., 2018a), energy (Ervural et al., 2018), construction risk assessment (Ergu et al., 2014; Liu and Tsai, 2012), and healthcare (Kuo et al., 2011; Feng et al., 2018), and it has been proven to be an effective methodology for decision making. Compared with another MCDM method, namely, the analytic hierarchy process (AHP), the ANP is more comprehensive and is a generalization of the AHP. The ANP provides a general framework to address decisions regarding the interdependence relationships among decision levels and attributes while the AHP does not allow for the measurement of the possible dependencies among factors (Yüksel and Dağdeviren, 2007).

Many decision problems cannot be structured hierarchically because they involve the interaction and dependence of higher-level elements on lower-level elements (Saaty, 2013). Therefore, unlike the tree-shaped hierarchical structure of the AHP, a network system should be constructed when using the ANP. The ANP divides the system elements into two hierarchies, one is the control hierarchy, which includes the problem objectives and decision criteria. All decision criteria in this hierarchy are considered to be independent and are only governed by the target, and the weight of each criterion in this hierarchy can be obtained by the AHP. The other hierarchy is the network hierarchy, in which dependencies and feedback within and between the elements can be considered. The network hierarchy is composed of all the elements controlled by the control hierarchy. The clusters of elements may influence each other, and the loops in the network indicate that the clusters have inner dependences. Thus, the network spreads out in all directions and its cluster of elements is not arranged in a particular order (Saaty, 2004b). The pairwise comparison is carried out for all the nodes of each cluster in relation to any other node of the network to represent the preference of decision makers, and the nine-point fundamental scale (Saaty, 1980) is used to express the judgements numerically. Thus, reciprocal comparison matrices consisting of judgements are constructed. The vector of priorities is the principal eigenvector of the comparison matrix, and these priorities are normalized to ensure that they sum to one and are unique. Similar to the AHP, consistency verification is perceived as one of the most significant strengths of the AHP (Ho, 2008), and it is also necessary and important in the ANP. The degree of consistency, *i.e.*, the consistency ratio (C.R.), among pairwise comparisons is a form of feedback for decision makers to modify their judgements. The derivation of C.R. requires the calculations of the maximum eigenvalue, consistency index (C.I.) and consistency ratio (C.R.). It is recommended that C.R. should be less than or equal to 0.10; otherwise an adjustment is needed to improve the consistency of the comparisons.

The priorities derived from pairwise comparison matrices are entered as parts of the columns of a supermatrix. To obtain the weighted supermatrix, each entry of the respective eigenvector for each column cluster needs to be multiplied by all the elements in the corresponding cluster of that column. The resulting matrix is then normalized so that the sum of all entries becomes one. Then the weighted supermatrix is raised to the power of $2k+1$, where k is an arbitrarily large number, to obtain the overall priorities; the new matrix is called the limit supermatrix (Saaty, 1996). The alternative with the largest overall priority is the best decision.

Sometimes, it would not be appropriate to use the ANP only to

solve all the decision-making problems since practical problems involve a variety of requirements and restrictions, and human judgements would be subjective. Thus, the integrations of the ANP with other MCDM methods, such as goal programming (GP) (Lee and Kim, 2000) technique for the order of preference by similarity to ideal solution (TOPSIS) (Shyur, 2006), the decision making trial and evaluation laboratory (DEMATEL) (Quezada et al., 2018), and the artificial neural network (ANN)(Alizadeh et al., 2018) provide more approaches to solve problems that are more complex. The fuzzy ANP is studied to address imprecise and uncertain human comparison judgements (Mikhailov and Singh, 2003). A generalized analytic network process (G-ANP) was developed to allow for multiple forms of preferences and was used to accomplish the risk assessment to the energy channels of China (Zhu et al., 2015). Further extension and improvement of the current work of the ANP will continue to increase the utility of the ANP.

There are some reviews related to the ANP, most of them focused on different decision-making problems and applied the ANP to determine the priorities of factors and alternatives. Energy is the most common topic among these reviews. An extensive literature review was conducted to identify the obstacles preventing the development of non-conventional renewable energy sources in Colombia by Gómez-Navarro and Ribó-Pérez (2018), and the ANP was adopted to obtain the assessment model and prioritize criteria and strategies. There is a review that integrated the ANP, fuzzy TOPSIS and SWOT (strengths, weaknesses, opportunities, and threats) analysis to formulate the energy strategy in Turkey after the presentation of a review of recent studies on energy planning and an overview of Turkeys energy planning (Ervural et al., 2018). Another hybrid methodology, the DEMATEL-ANP approach, was applied to construction risk assessment of oil and gas projects, and the possible risk factors were extracted from the literature (Dehdasht et al., 2017). Reviews on other topics, such as intelligent building systems (IBSs) (Wong et al., 2008), low-cost carriers (LCCs) (Lin and Huang, 2015) and reverse logistics (Ravi et al., 2005), were also conducted in a similar manner to the reviews mentioned above. A study that reported the applications of the ANP in quality function deployment (QFD) was briefly introduced by Sivasamy et al. (2016). Studies on the pairwise comparison matrix (PCM) (Kou et al., 2016) which has been widely used in the ANP, were classified and analyzed to find the hot topics and research techniques in the PCM.

In contrast to the above reviews, bibliometric techniques were adopted in this paper. Bibliometrics was defined as "the application of mathematics and statistical methods to books and other media of communication" by Pritchard (1969); quantitative methods are used in bibliometrics to make pronouncements about qualitative features of scientific studies (Wallin, 2005). Unlike traditional narrative reviews, which always rely on the experience and knowledge of researchers, bibliometrics examines science as a knowledge-generating system (van Raan, 2005) and provides a perspective that can easily be scaled from micro to macro level (Wallin, 2005). Bibliometrics is slightly different from the Preferred Reporting Items for Systematic Reviews and Meta-analyses(PRISMA) (Moher et al., 2009), although both involve more objective analysis methods, which evaluate papers by analyzing data provided from the database more quantitatively and qualitatively. Traditional bibliometric indicators include leading research areas, productive institutions, journals, authors, and corresponding publication volume and citation analysis. To better provide the trend of present or future study, author keyword analysis, especially the distribution change of author keywords in different periods was applied to reflect research trends in recent years (Chen et al., 2014, 2016b). Bibliometric techniques can be used as a tool for analyzing research activities and figuring out future research directions in different fields such as the AHP and TOPSIS techniques (Zyoud and Fuchs-Hanusch, 2017), water quality (Wang et al., 2016), organic chemistry (Wu et al., 2018), energy (Mao et al., 2015), food science (Yeung et al., 2018), soft robotics (Bao et al., 2018),deep learning (Faust et al., 2018), environmental monitoring (Zhang et al., 2017), manufacturing (Marzi et al., 2017), operations research and management science (Merigó and Yang, 2017).

Global trends and the development status quo of the ANP were evaluated in this paper by revealing the following aspects: global contributions, leading countries and regions, most productive institutions, journals, authors, leading research areas, collaboration patterns between countries/regions and institutions, highly cited articles and historical maps of author keywords, hot papers and hot topics.

The remainder of this paper is organized as follows. Section 2 describes the data source, search strategy and analysis methodology. Section 3 provides the results of the study and a discussion of these results. Section 4 gives the implications. Conclusions and limitations are presented in Section 5.

2. Data collection and analysis methods

The flow chart of the methodology used in this paper is presented in Fig. 1. The bibliometric analysis process can be divided into four parts: determination of the search query, data collection, data analysis and data visualization.

2.1. Paper collection

Paper collection was achieved through the Web of Science (WoS) database, which was used to retrieve documents related to the ANP. WoS is widely regarded as the standard and most widely used tools to generate citation data for scientific research and other assessment purposes (Meho and Yang, 2007). The WoS Core Collection of three Citation Indexes, namely, the Science Citation Index Expanded (SCIE), Social Science Citation Index (SSCI) and Art and Humanity Citation Index (A&HCI), includes more than 12,000 authoritative, high-impact academic journals in the world, including those in the natural sciences, engineering, biomedicine, social sciences, arts and humanities.[1] Literature were retrieved on August 2018, and the document types were refined for articles and reviews. Since the analytic network process(ANP) was proposed by consensus by Satty, T. L. in 1996, the year of publication was limited from 1996 to 2018.

The search query is determined by the expressions of the topic. Several possible expressions of the ANP in the WoS database advance search were used in the title, abstract and keywords to obtain the research output: analytic network process, analytical network process, analysis network process, analytical network procedure and analytic network approach. Thus, the search query was: TS= ("analy* network proce*" OR "analy* network approach"). As a result, 1338 documents were collected. The abbreviation "ANP" should also be used as the search term since some articles may only use the abbreviation in the title, abstract and keywords. However, when entering the query TS=("ANP" NOT "analy* network proce*" NOT "analy* network approach"), some vocabularies that are irrelevant but with the same expression were included in the search result. To obtain the documents that are associated with the ANP, we roughly concluded some unrelated full expressions and excluded them by using the Boolean operator "NOT" in the query to narrow the search range and then manually selected the relevant documents on the webpage. The exclusion and inclusion criteria and the criteria explanations of publications retrieved by the abbreviation are shown in

[1] https://clarivate.com/essays/journal-selection-process/.

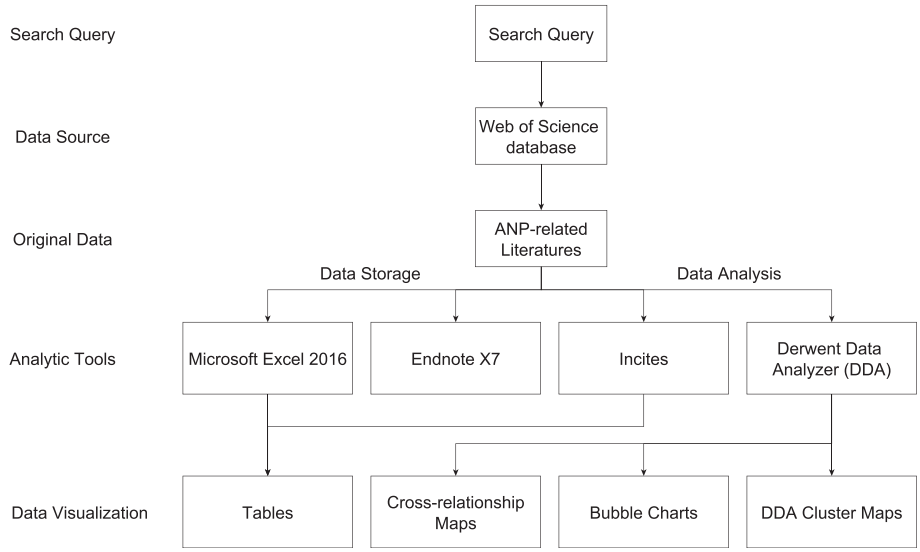

Fig. 1. The flow chart of the methodology.

Table 1. Finally, 1485 documents in total were obtained.

2.2. Data collection

Full record and cited references of all the retrieved documents were exported from Web of Science to Endnote and Derwent Data Analyzer. Endnote is the industry standard software tool for publishing and managing bibliographies, citations and references, and all the information of each paper can be stored in Endnote.

Derwent Data Analyzer (DDA) was used to manage the data extracted from the 1485 documents. DDA is a desktop-based data mining and visualization platform. It was applied for data cleaning, data mining and data visualization. It can analyze the number of papers in different WoS research areas, journals, countries/regions, institutions and authors by year. DDA can be used to count the frequency of keywords that were extracted from publications. In addition, various charts including cross relationship maps, bubble charts and cluster maps which can indicate the useful information behind data were generated through DDA.

Other indicators, such as the *h*-index, impact factor and number of citations, were analyzed using InCites, which is an analytical evaluation database based on Web of Science citation data, data cleansing and organization name normalization processing.

2.3. Analysis methods

Characteristics and trends of ANP research were quantitatively and qualitatively analyzed through bibliometric methods. Data collected from Web of Science were organized and visualized in tables and charts: line charts were used to illustrate the trend of global contributions; bibliometric indicators including the number of publications, number of citations, *h*-index and impact factor were presented in tables to show the characteristics of ANP research from different aspects; cross relationship maps and DDA cluster maps were applied to explain the collaborative relationships between research areas, countries/regions or institutions; and bubble charts were adopted to show the development trends of research areas, journals, author keywords and authors in ANP research more intuitively.

Hot topics can provide information about the research interests and perspectives of the year, so each year's hot papers should be organized. The Pareto principle and the data derived from the citation report downloaded from Web of Science were used to sort out the hot papers of each year, that is, first, the papers published in the last two years of each year are selected (e.g., papers published in 2016 and in 2017 are the candidate hot papers in 2017). Second, the top 10% (which can be adjusted according to the amount of literature) of papers with high citations are chosen. The ratio of hot papers is set to 10% when the number of studies is not less than 20; otherwise, it is set to 20%. Hot topics were concluded from the author keywords of the hot papers.

3. Results and discussions

As mentioned in Section 2, 1485 documents including 1442 articles and 43 reviews were obtained, and 70 countries have

Table 1
Exclusion and Inclusion criteria of publications retrieved by the abbreviation.

Exlusion/ Inclusion	Criteria	Criteria Explanation
Exclusion	TU: Totally Unrelated	TU-1: There are full expressions of “ANP” in the title, abstract, author keywords or keywords plus, but they are totally not the expressions of the same meaning as analytic network process. TU-2: There are only abbreviations “ANP” in the title, abstract, author keywords or keywords plus, but they are only the names of other unrelated items (e.g., chemical, medical or biological items).
Inclusion	PR: Partially Related	PR-1: There are abbreviations “ANP” only appear in the keywords plus rather than in the title, abstract or author keywords, and the literature is related to analytic hierarchy process or multi-criteria decision making. PR-2: There are not any abbreviations (“ANP”) in the title, abstract, author keywords or keywords plus, but analytic network process is mentioned in the literature.
	TR: Totally Related	TR: There are only abbreviations (“ANP”) in the title, abstract, author keywords or keywords plus, and analytic network process is the main content in the literature.

contributed to the ANP research field. The growth trend of the published research over time for the ANP is shown in Fig. 2. The growth rate has been relatively stable in the first thirteen years, and a steep rise occurred in 2009, with a high rate of increase of 81.58%. The number of documents remained approximately 180 between 2015 and 2017. It is calculated that more than 90% of the literature was published during the period from 2009 to 2018. Although the ANP was first proposed in 1996, the first article retrieved from the WoS database was published in 1997 by Meade et al. (1997) as the first author. In this article, a structured approach and strategic analysis techniques, including the use of the ANP and utility theory, were presented to evaluate strategic alliances and help reduce the risk of alliance formation.

3.1. Contribution of leading research areas

Research areas reflect application scopes of the topic. These 1485 papers on the ANP cover 107 WoS categories. The Web of Science category/research area is one of the information contained in each publication, and this category is sorted by Web of Science. Table 2 illustrates the top 20 WoS research areas ranked by the total number of articles related to the ANP. “Operations research and management science” dominates the research area list with 373 articles, followed by “computer science, artificial intelligence”, “engineering, industrial”, “environmental sciences”, “management” and “Engineering, Manufacturing”. The remaining research areas contributed a share of less than ten percent. “engineering, environmental” and “engineering, electrical and electronic” lead the list of average citations per publication (ACPP), with ACPP of 30.72 and 30.16, respectively. The results show that the ANP is commonly applied in operations research and is an effective tool to solve management and engineering problems.

The emergence of cross-relationships between research areas can be attributed to the development of theoretical methods and the diversity of practical problems. The relationships between the top 20 research areas are given in Fig. 3. The size of nodes of the cross-relationship maps is proportional to the total number of publications of each WoS category. Some publications can have multiple Web of Science categories, so the WoS categories that are labelled on the same papers will be connected; the greater the frequency with which they appear together in publications, the stronger the collaborative relationship is. Therefore, the lines of cross-relationship maps represent the collaboration between research areas; the stronger the collaboration is, the thicker the lines are.

Articles belonging to the “operations research and management science” research area are usually categorized into “computer science, artificial intelligence”, “engineering, manufacturing” and “engineering, electrical and electronic”. “environmental sciences” has intensive relationships with the WoS research areas related to the environment, such as “environmental studies”, “engineering, environmental” and “green and sustainable science and technology”.

The development trend of a certain aspect can be presented through a bubble chart. As shown in Fig. 4, the numbers on the top of the chart are publishing years. The number in each bubble is the annual number of publications belonging to the WoS category listed on the left-most side of the chart. The larger the bubble is, the more publications there are. By comparing the sizes of bubbles horizontally, the growth trend of each research area over time can be determined; by comparing the sizes of the bubbles vertically, the hot research areas of each year can be identified. The first ANP-related article in the “operations research and management science” research area was published in 1997, and the number of articles in this research area dramatically increased in 2009 and reached its highest value in 2015 with 46 articles. In addition, there have been more than 20 articles for each year from 2009 to 2017. The application of the ANP in the “computer science, artificial intelligence” research area started in 2003, and the number of articles soared to 26 in 2009 and peaked at 40 in 2011. For the “engineering, industrial” research area, the years from 2009 to 2015 witnessed a relatively steady rise in the number of articles despite of some fluctuations during this period. In recent three years (2016–2018), “operations research and management science” is no longer the most dominant research area, and the number of articles in “environmental sciences”, “green and sustainable science and technology” and “environmental studies” has significantly increased and all reached their peak values in 2017. The“Engineering, Electrical & Electronic” research area included most of the publications related to the ANP during 2009 and 2011 (24, 19 and 28

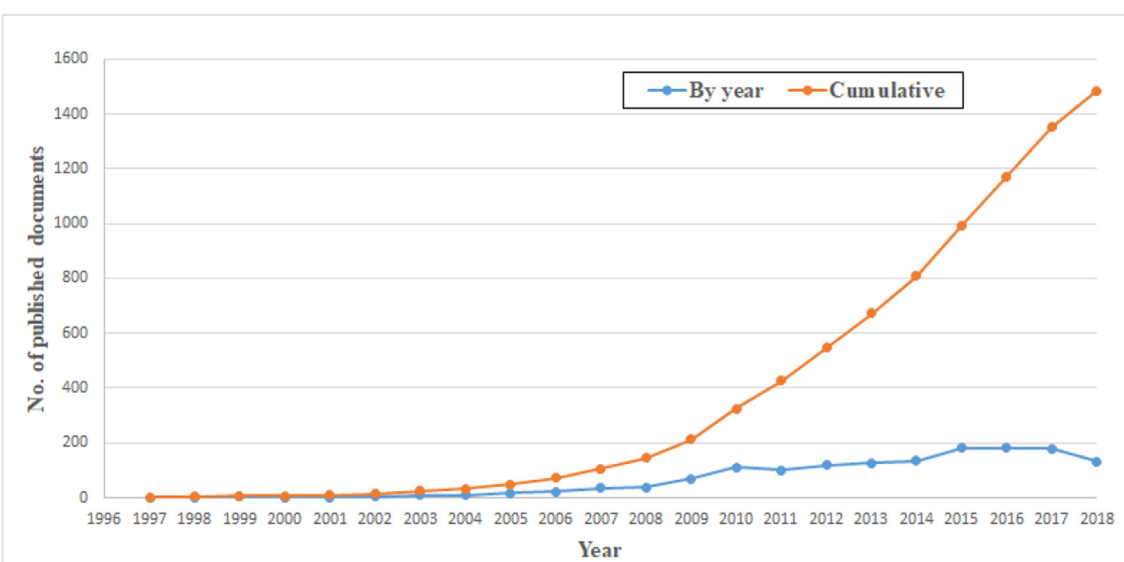

Fig. 2. Number of published documents from the global related to ANP.

Table 2
Contribution of the top 20 WoS research areas in ANP research.

Rank	WoS research area	TA	TPR%	TC	ACPP
1	Operations Research & Management Science	373	25.12	10654	28.56
2	Computer Science, Artificial Intelligence	262	17.64	6558	25.03
3	Engineering, Industrial	202	13.60	5208	25.78
4	Environmental Sciences	188	12.66	3541	18.84
5	Management	172	11.58	4488	26.09
6	Engineering, Manufacturing	158	10.64	3193	20.21
7	Computer Science, Interdisciplinary Applications	141	9.49	3748	26.58
8	Engineering, Electrical & Electronic	139	9.36	4192	30.16
9	Green & Sustainable Science & Technology	130	8.75	2358	18.35
10	Environmental Studies	89	5.99	848	9.53
11	Engineering, Civil	78	5.25	1080	13.85
12	Engineering, Environmental	75	5.05	2304	30.72
13	Engineering, Multidisciplinary	73	4.92	981	13.44
14	Business	71	4.78	1173	16.52
15	Economics	70	4.71	931	13.30
16	Computer Science, Information Systems	68	4.58	1058	15.56
17	Energy & Fuels	64	4.31	1102	17.22
18	Mathematics, Interdisciplinary Applications	47	3.16	648	13.79
19	Mathematics, Applied	40	2.69	904	22.60
19	Automation & Control Systems	40	2.69	459	11.48

Abbreviations: TA = total articles; TPR% = the percentage of articles of journals in total publications; TC = total citations; ACPP = average citations per publication.

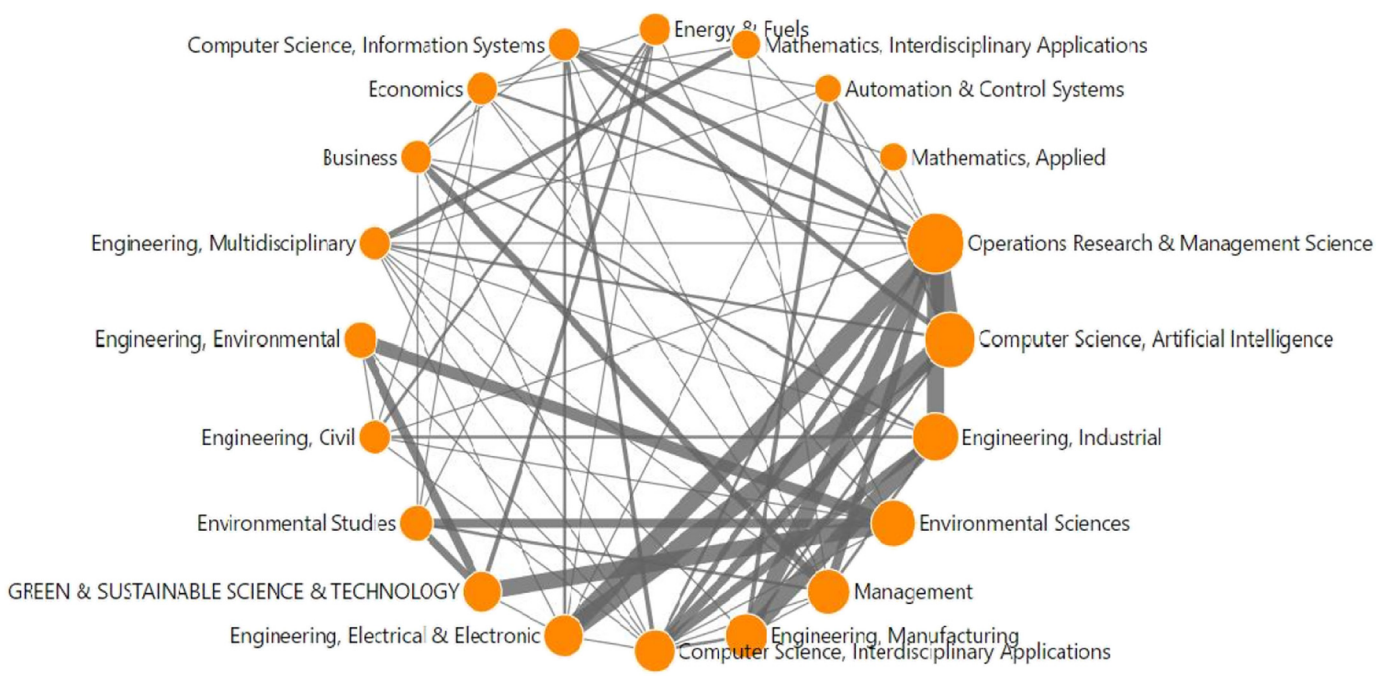

Fig. 3. The cross relationship map of the top 20 WoS research areas.

articles for each year, respectively), and fell to 4 in 2017.

3.2. Leading journals that have contributed to the ANP

It is essential for scholars and those who are interested in the ANP to know about the journals that publish ANP research and decide which journal to read or to which journal to submit their research papers. In the ANP field of research, 397 journals contributed to the ANP literature. As listed in Table 3, *Expert Systems with Applications* is the dominant source of ANP research with 118 (7.95%) documents in total and the highest number of total citations (4,096). It is followed by *Journal of Cleaner Production* (47; 3.26%) and *Sustainability* (45; 3.03%); but the average citation per publication of *Sustainability* is the second-lowest on the list. This placement may be attributed to *Sustainability* publishing an article related to the ANP for the first time in 2014, after 19 other journals had done so, but its contribution is still considerable. The ACPP of the *European Journal of Operational Research* is the highest (57.52), followed by *Information Sciences* (44.83), which has fewer articles (12; 0.81%) about the ANP in the list. In terms of impact factor (IF), *Renewable & Sustainable Energy Reviews* has the highest value of 9.184, and the *Journal of Cleaner Production* takes second place with 5.651.

To show the trend of the top 20 most productive journals by year, a bubble chart was employed, as shown in Fig. 5. During the period from 1998 to 2006, few articles were sparsely distributed among the top 20 journals. Journals including the *International Journal of Production Research*, the *International Journal of Advanced*

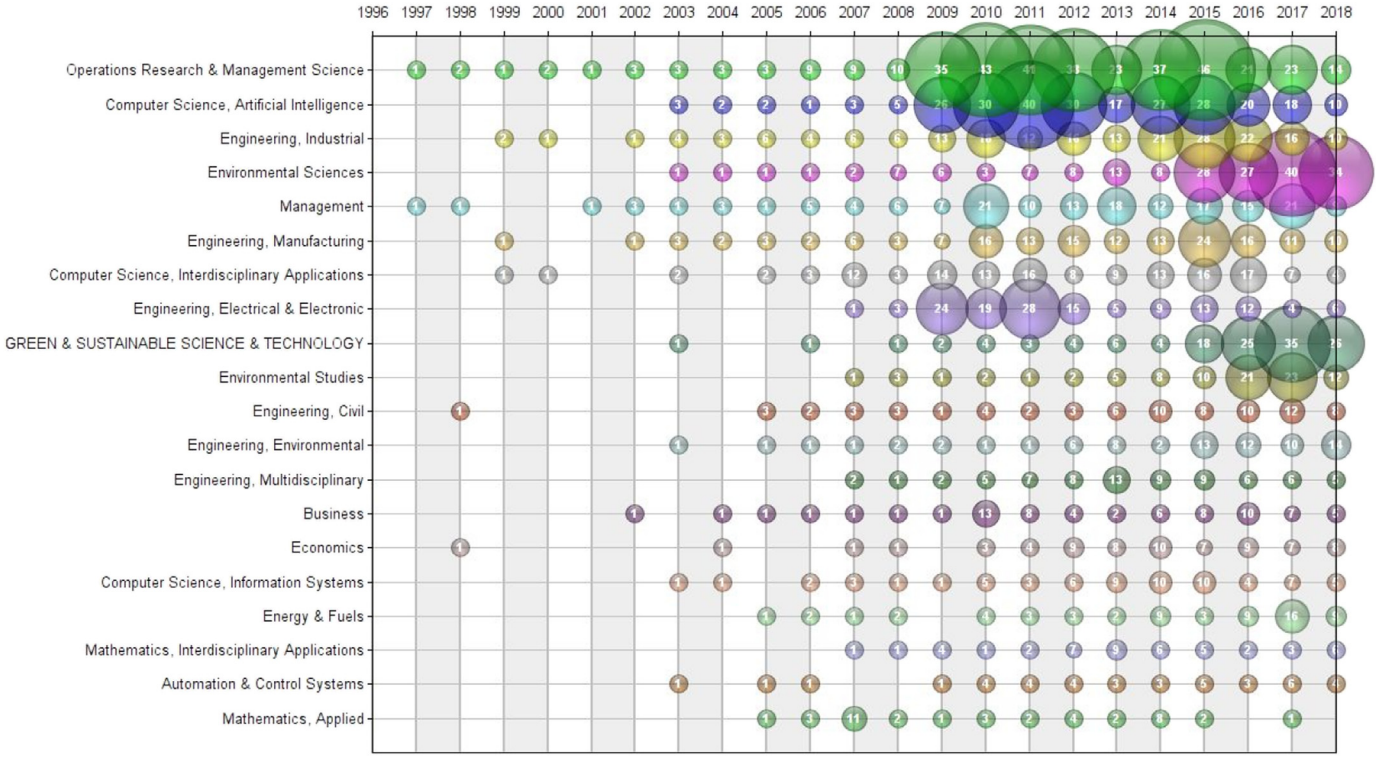

Fig. 4. Bubble chart of top 20 WoS research areas by year.

Table 3
The top 20 productive journals publishing articles in ANP research.

Rank	Journal	TA	TC	ACPP	IF
1	Expert Systems with Applications	118	4096	34.17	3.768
2	Journal of Cleaner Production	47	1672	35.57	5.651
3	Sustainability	45	203	4.51	2.075
4	International Journal of Production Research	41	1189	29.00	2.623
5	Computers & Industrial Engineering	38	1431	37.66	3.195
6	International Journal of Production Economics	33	845	25.61	4.407
7	European Journal of Operational Research	27	1553	57.52	3.428
8	Mathematical Problems inEngineering	25	139	5.56	1.145
9	International Journal of Advanced Manufacturing Technology	22	265	12.05	2.601
10	Journal of Intelligent & Fuzzy Systems	20	137	6.85	1.426
11	Journal of Testing and Evaluation	20	37	1.85	0.669
12	Applied Soft Computing	19	402	21.16	3.907
13	International Journal of Information Technology & Decision Making	17	101	5.94	1.755
14	Mathematical and Computer Modelling	15	526	35.07	1.366[a]
15	Production Planning & Control	15	239	15.93	2.33
16	Technology and Economic Development of Economy	15	209	13.93	3.244
17	Journal of the Operational Research Society	14	225	16.07	1.396
18	Renewable & Sustainable Energy Reviews	13	309	23.77	9.184
19	Safety Science	13	246	18.92	2.835
20	Information Sciences	12	538	44.83	4.305

Abbreviations: TA = total articles; TC = total citations; ACPP = average citations per publication; IF = impact factor.
[a] The impact factor was extracted from the 2015 edition of Journal Citation Reports (JCR) since Mathematical and Computer Modelling is discontinued as of 2014.

Manufacturing Technology, the *Applied Soft Computing*, and the *International Journal of Information Technology & Decision Making* have relatively stable annual publication volumes on ANP research. Most of the articles in ANP research published in *Expert Systems with Applications* were published between 2009 and 2012, and suddenly dropped to 4 in 2013; thereafter, the number of articles in subsequent years was no more than 10. The development trend of the *Journal of Cleaner Production* is the opposite of that of *Expert Systems with Applications*; the number of articles increased notably from 2 in 2014 to 8 in 2015 and remained at approximately 10 during recent four years.

The top 3 most productive journals were selected to analyze the hotspots of ANP research in different journals. In the articles published by *Expert Systems with Applications*, different MCDM methods were integrated with the ANP or the fuzzy ANP for solving various decision making problems, including, but not limited to, DEMATEL, ZOGP (zero-one goal programming) and the ANP were combined to evaluate sourcing decisions (Tsai et al., 2010). DEMATEL, VIKOR (Visekriterijumska Optimizacija I Kompromisno Resenje) and the ANP were hybridized to select outsourcing

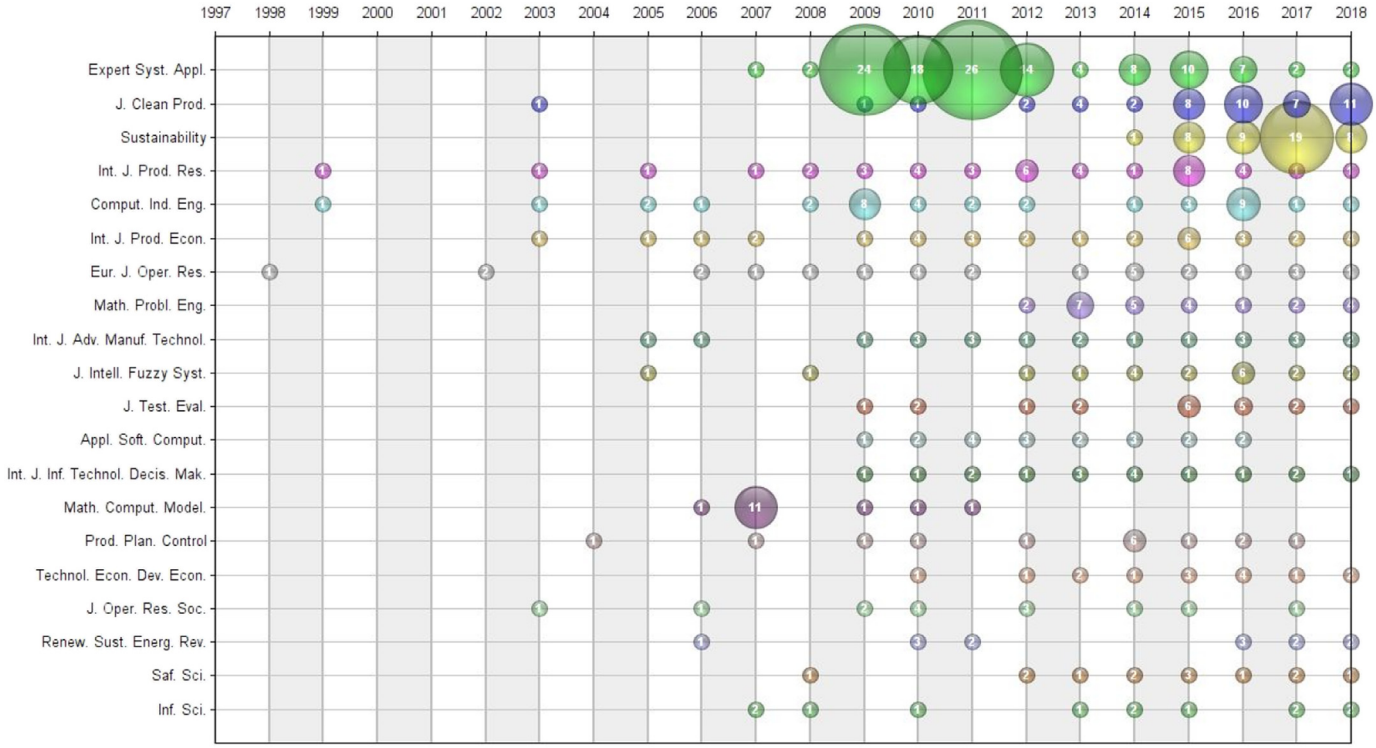

Fig. 5. Bubble chart of top 20 productive journals by year.

providers (Liou and Chuang, 2010); the hybrid approach based on the fuzzy ANP, fuzzy DEMATEL and fuzzy VIKOR was applied to the city logistics (CL) concept selection (Tadić et al., 2014). Personnel selection could be conducted through a combination of the fuzzy ANP, fuzzy TOPSIS and fuzzy ELECTRE (ELimination Et Choix Traduisant la REalit) techniques (Kabak et al., 2012). Supplier selection and supply chain management are the most popular topics of *Expert Systems with Applications*, including the outsourcing vendor selection (Lin et al., 2010), green supply chain management (Tseng et al., 2014), divergent supply chain (DSC) planning (Hung, 2011), closed-loop supply chain configuration (Tosarkani and Amin, 2018) and other topics.

Supplier selection and supply chain management are also hot topics of the *Journal of Cleaner Production*. The top 3 most highly cited articles published in this journal are all related to supplier selection or supply chain management. Each of these articles has gained more than 170 citations, and as shown in Table 7, two of these three articles (Sarkis, 2003; Kuo et al., 2010) were listed as the top 20 most cited documents, ranking first and sixteenth with 513 and 195 citations, respectively. Sarkis (2003) discussed the components and elements of green supply chain management that were used to develop the decision framework. Kuo et al. (2010) hybridized the artificial neural network (ANN), the data envelopment analysis (DEA) and the analytic network process (ANP) to evaluate green suppliers' performances. Hsu and Hu (2009) applied the ANP to supplier selection which incorporated the issue of hazardous substance management (HSM), and this article has been cited 175 times. In addition to these top three articles, other articles that published in recent years also applied the ANP to the supply chain. Rajesh and Ravi (2015) compared the supplier alternatives obtained by the methods of the AHP, the ANP and Grey relational analysis in the context of a resilient supply chain; Wu and Barnes (2016) used an integrated approach combining the ANP and multi-objective programming (MOP) methods to construct a model for green partner selection and supply chain.

For the *Sustainability*, topics that have been continuously focused on are green supplier selection (Hsu et al., 2014; Kuo et al., 2015; Chung et al., 2016; Lin et al., 2018) and location selection (Yang et al., 2015; Gigović et al., 2016; Lee et al., 2017).

3.3. Analysis of author keywords

To reveal the main focus and research trend of ANP research, 3041 author keywords from the retrieved results were analyzed. The keywords with the same meanings were unified into one word, and publications that lack author keywords may not be included in this analysis. Research trends may be more obvious with the analysis of the distribution change of author keywords over different lengths of time, but the most concering recent issues probably cannot be reflected, and trends would be difficult to determine (Chen et al., 2016b). These problems can be settled by applying bubble charts. Bubble chart displays three dimensions of data, *i.e.*, author keywords, year and number of publications. Horizontal comparisons of the sizes of the bubbles can be used to determine the growth trend of each author keyword over time; vertical comparisons of the sizes of the bubbles can be used to identify the hot keywords of each year.

There were 2459 keywords that were used only once, accounting for 80.86% of the total, which indicates a wide range of interests in ANP research. As shown in Fig. 6, except for "ANP", which has been used 807 times in total, "MCDM" (266 times) is the most-used keyword and has increased dramatically between 2011 and 2016. It is followed by "DEMATEL", "fuzzy ANP" and "AHP", which have been used 164, 138 and 128 times, respectively. "AHP" has been used in ANP research since 1998, which indicates that the AHP is indeed the basic form of ANP research. "DEMATEL" was first applied to ANP research in 2008, Wu (2008) proposed a solution combining the ANP and DEMATEL to evaluate and select knowledge management strategies, since DEMATEL can be used to handle the

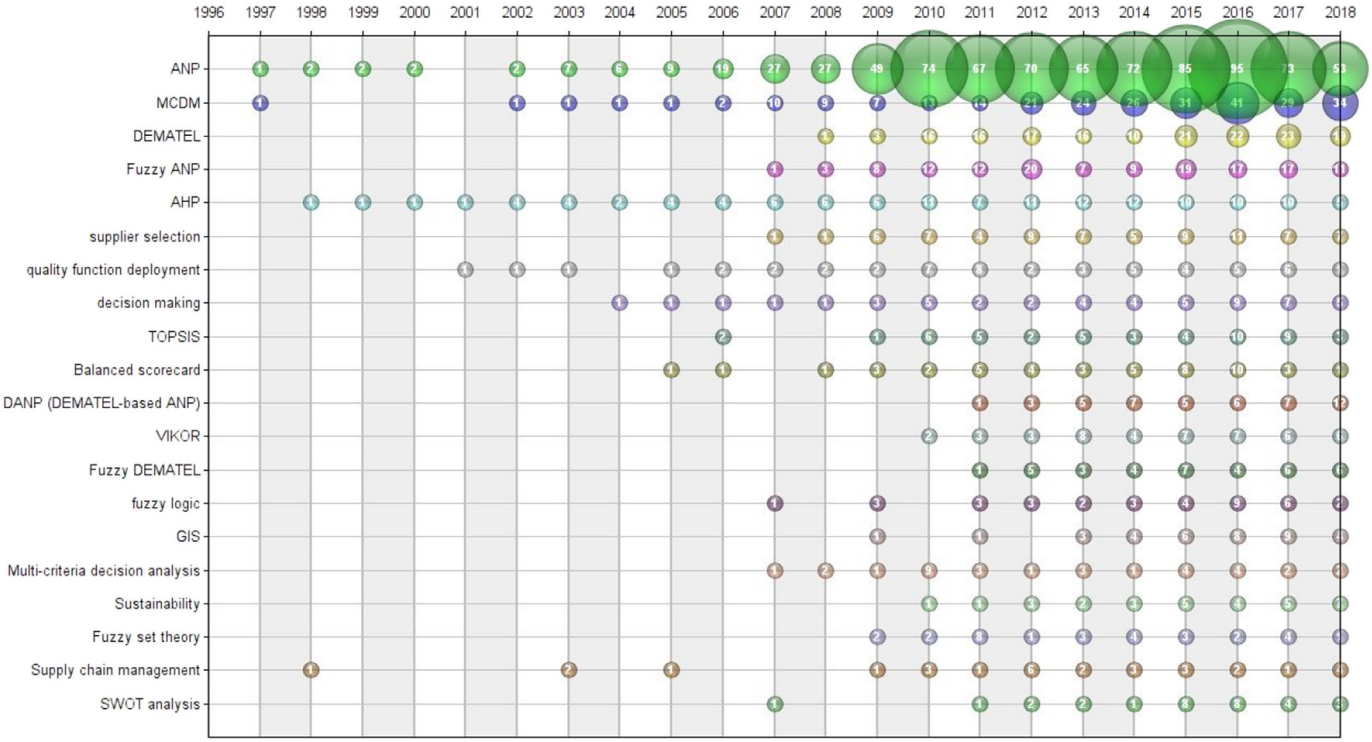

Fig. 6. Bubble chart of top 20 author keywords by year.

inner dependences within an evaluation cluster. From then on, DEMATEL plays an important role in constructing interrelations among criteria before applying the ANP to obtain criteria weights, and its frequency of use has continued to grow recent years. "DANP (DEMATEL-based ANP)" (46 times) has been used as a keyword since 2011 and, similar to "DEMATEL", exhibits a growing trend.

"Supplier selection" is ranked sixth (74 times in total), and other keywords that are related to the supply chain, such as "supply chain management", "supplier evaluation", "green supply chain management", "green supply chain (GSC)", "sustainable supply chain management" and "green supplier selection", have also attracted scholars' interests. Supplier selection has been one of the hot topics in ANP research since 2007 (Gencer and Gürpinar, 2007). The ANP was first applied to supply chain management in 1998, and Meade (1998) established a systemic analytical model that could be utilized to evaluate and make optimal logistics strategies in the dynamic environment. With the increasing awareness of environmental issues and sustainable development, green indicators have been added to the supplier evaluation process and supply chain management since 2003 (Sarkis, 2003), and studies on green supplier selection and the green supply chain have gradually developed.

"Quality function deployment" is in the seventh place (55 times) and appeared in ANP research in 2001 when Partovi (2001) used the ANP to determine the intensity of synergy effects among column variables in quality function deployment (QFD) matrices. Partovi and Corredoira (2002) also combined the ANP and QFD to improve the game of soccer to make it more attractive. Approaches that combine the ANP and QFD were applied to manufacturing, such as product planning (Karsak et al., 2003), facility location selection (Partovi, 2006), and rapid tooling route selection (Pal et al., 2007); supply chain management has also been a main application area since 2011 (Büyüközkan and Berkol, 2011), and sustainable supply chain has been particularly studied in the past five years (Zhang and Awasthi, 2014; Lam and Dai, 2015; Lam, 2015). Fuzzy QFD which incorporates the fuzzy ANP and QFD was developed to determine the internal relationships among factors in the house of quality (HoS) under the fuzzy environment (Güngör et al., 2011; Lee and Lin, 2011; Zaim et al., 2014).

The use of the "balanced scorecard" in ANP research was first presented in 2005 and reached its highest value in 2016 (10 times). In 2005, Ravi et al. (2005) employed a combination of a balanced scorecard and ANP-based approaches to evaluate the alternatives for end-of-life (EOL) computers. A variety of MCDM methods, including "TOPSIS", "VIKOR (Visekriterijumska Optimizacija I Kompromisno Resenje)", "zero-one goal programming (ZOGP)", "data envelopment analysis (DEA)", "PROMETHEE (preference ranking and organization method for enrichment evaluation)" and "goal programming" were integrated with the ANP, and these integrated methods were used at a relatively stable frequency every year. Except for these MCDM techniques, the ANP was also combined with "GIS (geographic information system)", and the usage frequency of "GIS" has been increasing in recent five years.

There are also many keywords with the word "fuzzy", such as "fuzzy ANP" (136 times) which took the fourth position, as well as "fuzzy DEMATEL", "fuzzy TOPSIS", "fuzzy VIKOR", and "fuzzy Delphi". This finding should be attributed to the emergence of "fuzzy logic" and "fuzzy set theory" (Zadeh, 1965), which can help improve the objectivity of the evaluation results and reduce the inaccuracy caused by the lack of clarity and unpredictable human judgement regarding preferences (Büyüközkan and Çifçi, 2012). Different new techniques, such as intuitionistic fuzzy sets (Pei and Zheng, 2012) and intuitionistic fuzzy variables (Pei, 2015), were developed based on fuzzy set theory to address multi-attribute decision-making (MADM) problems. The fuzzy ANP has been applied to supplier selection (Büyüközkan and Çifçi, 2011), hospital information system (Nilashi et al., 2016), agriculture (Demirel et al., 2012) and other fields.

3.4. Global contribution and leading countries/regions

Analysis of academic research results from the national or regional level can indicate which countries/regions have more attention in a certain discipline. Collaborative relationships among countries/regions can help researchers or learners find appropriate collaborators or countries for further study. The top 20 countries in terms of the number of publications related to the ANP are shown in Table 4. The collaborative relationships of the top 20 most productive countries/regions are shown in Fig. 7. Articles originating from England, Scotland, Northern Ireland and Wales were grouped under the United Kingdom heading.

Taiwan is the most productive region with a total of 436 publications (29.36%) since 2004, followed by the People's Republic of China with 230 publications. Turkey is ranked third with 218 publications in total, followed by Iran (165 publications; 11.11%) and the United States (142 publications; 9.56%). Lithuania is the leading country in terms of joint publications with other countries, and a large number (94.29%) of its publications are international articles with a total of 35 published articles since 2011. The contributions related to ANP research in Australia, Singapore and Denmark are similar to those in Lithuania, with the feature of a relatively small number of articles but high cooperation rates. It should be noted that the average citation per publication (ACPP) of Denmark is quite high (47.86) despite its low publication volume. The People's Republic of China and the United States also have strong collaboration with other countries, with high share of publications of 48.26% and 68.31%, respectively. All of these high percentages of international articles indicate that the ANP has attracted worldwide scholars to exchange great ideas and cooperate with each other. In terms of collaborative relationships, as is shown in Fig. 7, the United States is the most active country that cooperated with other countries, especially with Iran, Turkey, China and Germany. China also has close relationships with Taiwan, the United States, the United Kingdom and Australia.

3.5. Contribution of leading institutions

The statistical analysis of research institutions provides more detailed and more specific information for scholars than does the analysis of countries/regions. Table 5 shows the top 20 most productive institutions in ANP research in terms of the total number of publications, citations and *h*-index. Most of them are from the top 10 most productive countries. A total of 60% of the top 20 institutions are from Taiwan, and National Chiao Tung University tops the list with the most publications, followed by Islamic Azad University and Kainan University. For the ACPP, Galatasaray University and Kainan University lead the list with 45.81 and 38.46, respectively, followed by Gazi University (37.86) and National Chiao Tung University (37.50). National Chiao Tung University and Kainan University are not only the top 2 institutions with respect to the total number of citations but also have the highest *h*-index values (28 and 26, respectively). These institutions play an important role in the development and promotion of the ANP research field.

The organization of collaborative relationships among institutions can help researchers find more opportunities to cooperate with other institutions. The collaboration network among the top 20 institutions is shown in Fig. 8. The numbers next to institution names represent the total number of publications from corresponding institutions. The number at each intersection refers to the number of collaborative articles with the other top 20 most productive institutions. Another number on each node indicates the number of publications resulting from the independent work of the institute or the collaboration with organizations other than the top 20 institutions. As shown in Fig. 8, institutions from Taiwan, including National Chiao Tung University, Kainan University, National Taipei University and National Taipei University of Technology, have established a large collaborative network, while Universitat Politecnica de Valencia, Gazi University and Hong Kong Polytechnic University did not cooperate with the other top 20 organizations.

3.6. Leading authors who contributed to the ANP

Leading authors who contributed to a certain area have a high reputation and their works can inspire scholars to identify research directions. Table 6 lists the top 20 most prolific authors based on the number of publications. All of these top 20 most productive authors are from the top 12 most productive countries. Tzeng, G. H. is the most productive author in fields related to ANP research with 56 (3.77%) publications. He is followed by the following authors: Lee, A. H. I. (29; 1.95%) from Chung Hua University, Taiwan; Kang, H. Y. (22; 1.48%) from National Chin-Yi University of Technology; and

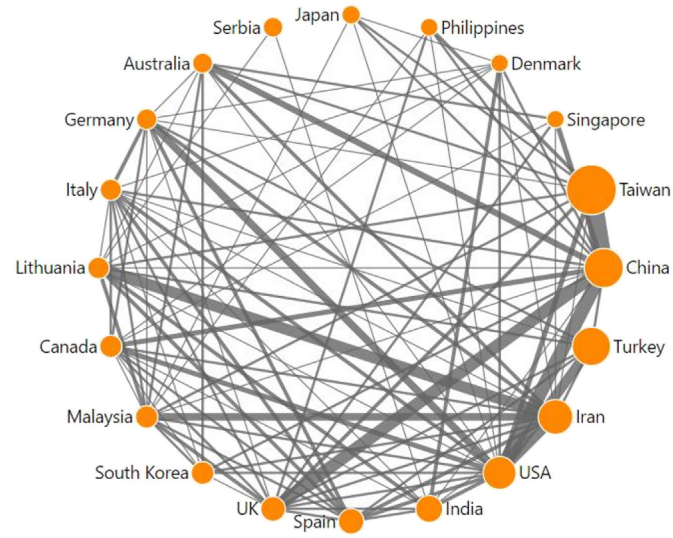

Fig. 7. Collaboration matrix map among the top 20 most productive countries/regions.

Table 4
The top 20 most productive countries/regions in ANP research.

Rank	Country/Region	TA	TC	ACPP	SP(%)
1	Taiwan	436	8260	18.94	14.22
2	People's Republic of China	230	3446	14.98	48.26
3	Turkey	218	5469	25.09	12.84
4	Iran	165	1733	10.50	40.61
5	United States	142	4470	31.48	68.31
6	India	76	1583	20.83	26.32
7	Spain	63	918	14.57	41.27
8	United Kingdom	56	1088	19.43	76.79
9	South Korea	42	676	16.10	26.19
10	Canada	40	583	14.58	67.50
11	Malaysia	40	486	12.15	75.00
12	Lithuania	35	571	16.31	94.29
13	Italy	33	561	17.00	51.52
14	Germany	29	577	19.90	68.97
15	Australia	26	468	18.00	88.46
16	Serbia	24	209	8.71	4.17
17	Japan	18	261	14.50	33.33
18	Philippines	15	262	17.47	60.00
19	Denmark	14	670	47.86	78.57
20	Singapore	14	164	11.71	85.71

Abbreviations: TA = total articles; TC = total citations; ACPP = average citations per publication; SP = share of publications.

Table 5
The top 20 most productive institutions in ANP research.

Rank	Institutions	TA	TPR%	TC	ACPP	*h*-index	Country/Region
1	National Chiao Tung University	60	4.04	2250	37.50	28	Taiwan
2	Islamic Azad University	59	3.97	618	10.47	13	Iran
3	Kainan University	48	3.23	1846	38.46	26	Taiwan
4	University of Tehran	45	3.03	583	12.96	13	Iran
5	National Taipei University	43	2.90	550	12.79	14	Taiwan
6	National Taipei University of Technology	42	2.83	1129	26.88	18	Taiwan
7	Universitat Politecnica de Valencia	36	2.42	557	15.47	12	Spain
8	Vilnius Gediminas Technical University	34	2.29	570	16.76	14	Lithuania
9	Chung Hua University	33	2.22	636	19.27	15	Taiwan
10	Galatasaray University	31	2.09	1420	45.81	14	Turkey
11	Istanbul Technical University	31	2.09	915	29.52	14	Turkey
12	Tamkang University	30	2.02	641	21.37	12	Taiwan
13	Gazi University	28	1.89	1060	37.86	14	Turkey
14	National Chin-Yi University of Technology	28	1.89	416	14.86	11	Taiwan
15	Chinese Culture University	27	1.82	144	5.33	6	Taiwan
16	Hong Kong Polytechnic University	26	1.75	600	23.08	16	Hong Kong
17	Ming Chuan University	26	1.75	238	9.15	9	Taiwan
18	National Taiwan University of Science and Technology	26	1.75	679	26.12	11	Taiwan
19	National Central University	25	1.68	623	24.92	13	Taiwan
20	National Cheng Kung University	20	1.35	337	16.85	12	Taiwan

Abbreviations: TA = total articles; TPR% = the percentage of articles of journals in total publications; TC = total citations; ACPP = average citations per publication.

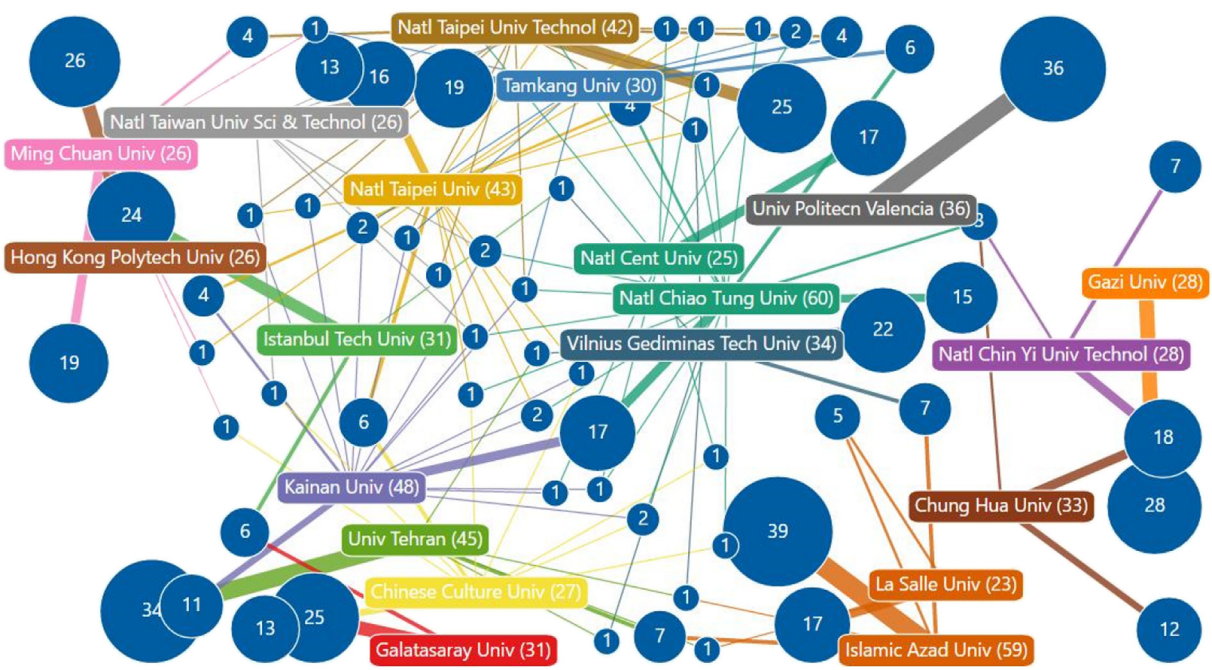

Fig. 8. DDA cluster map on cooperation of top 20 institutions.

Sarkis, J (22; 1.48%) from Clark University and Worcester Polytechnic Institute. Tzeng, G. H. also achieves the highest *h*-index of 26, followed by Sarkis, Joseph (19) and Lee, A. H. I. (15). These prolific authors also collaborate with each other. Tzeng, G. H. has cooperated with Hu, S. K. and Liou, J. J. H., completing 6 and 5 articles, respectively. Lee, A. H. I. has completed 20 articles with Kang, H. Y.; several application fields were involved in these 20 articles, such as new product development (NPD) (Chen et al., 2008; Lee et al., 2010), renewable energy (Kang et al., 2011; Lee et al., 2012) and supplier selection (Kang et al., 2012).

The change in a researcher's attention to a topic can be represented through the bubble chart. The bubble chart of the top 20 most productive authors by year is shown in Fig. 9. Sarkis, J. is the author who earliest published the paper related to the ANP in 1997 and is among the top 20 authors; he employed the ANP to support the strategic alliance management process (Meade et al., 1997). The number of articles produced by these top 20 authors reached their peak in 2010 with a total of 50, followed by 2017 and 2011 with 45 and 42 articles, respectively. From 2011 to 2015, the number of articles authored by Tzeng, G. H. remained the highest. The articles authored by Tzeng, G. H. cover a wide range of topics, such as education (Chen and Tzeng, 2011; Hu et al., 2018), investment strategies (Ho et al., 2011), and marketing strategies (Wang and Tzeng, 2012), and the topics of the articles of the last three years have focused on mobile commerce (Hu et al., 2015; Lu et al., 2015), the digital economy (Hsu et al., 2018), and sustainable development (Liu et al., 2018a).

Table 6
Contribution of the top 20 most productive authors in ANP research.

Rank	Author	TA	TPR%	h-index	Affiliation
1	Tzeng, G. H.	56	3.77	26	National Chiao Tung University, Taiwan National Taipei University, Taiwan Nan Kai University Technology, Taiwan.
2	Lee, A. H. I.	29	1.95	15	Chung Hua University, Taiwan.
3	Kang, H. Y.	22	1.48	10	National Chin-Yi University of Technology, Taiwan.
3	Sarkis, J.	22	1.48	19	Clark University, United States; Worcester Polytechnic Institute, United States.
5	Tseng, M. L.	21	1.41	13	Lunghwa University of Science and Technology, Taiwan; MingDao University, Taiwan.
6	Zavadskas, E. K.	18	1.21	13	Vilnius Gediminas Technical University, Lithuania.
6	García-Melón, M	18	1.21	10	Universitat Politecnica de Valencia, Spain.
6	Liou, J. J. H.	18	1.21	12	National Taipei University of Technology; Kainan University, Taiwan.
9	Büyüközkan, G.	17	1.14	13	Galatasaray University, Turkey.
9	Dağdeviren, M.	17	1.14	10	Gazi University, Turkey.
11	Chen, H. H.	16	1.08	7	Macau University of Science & Technology, China; Tsinghua University, China.
12	Lin, C. T.	15	1.01	9	Ming Chuan University, Taiwan; Yuanpei University, Taiwan.
13	Tsai, W. H.	14	0.94	10	National Central University, Taiwan.
14	Aragonés-Beltrán, P.	13	0.88	10	Universitat Politecnica de Valencia, Spain.
14	Kabak, M.	13	0.88	5	Gazi University, Turkey; Turkish Military Academy, Turkey; Kara Harp Okulu Komutanligi, Turkey.
16	Kahraman, C.	11	0.74	9	Istanbul Technical University, Turkey.
16	Tavana, M.	11	0.74	7	University of Paderborn, Germany; La Salle University, United States.
18	Chen, I. S.	10	0.67	6	National Chiao Tung University, Taiwan; Trinity College Dublin, Ireland; Wuhan University, China.
18	Wey, W. M.	10	0.67	5	National Taipei University, Taiwan; National United University, Taiwan.
20	Hu, S. K.	9	0.61	5	Kainan University, Taiwan.
20	Zayed, T.	9	0.61	2	Concordia University, Canada.

Abbreviations: TA = total articles; TPR% = the percentage of articles of journals in total publications.

Table 7
The top 20 most cited publications related to ANP research.

Rank	Author, (PY)	Title	Journal	TC	TCY
1	Sarkis (2003)	A strategic decision framework for green supply chain management	Journal of Cleaner Production	513	32.06
2	Kahraman et al. (2006)	A fuzzy optimization model for QFD planning process using analytic network approach	European Journal of Operational Research	307	23.62
3	Lee and Kim (2000)	Using analytic network process and goal programming for interdependent information system project selection	Computers & Operations Research	294	15.47
4	Karsak et al. (2003)	Product planning in quality function deployment using a combined analytic network process and goal programming approach	Computers & Industrial Engineering	286	17.88
5	Meade and Sarkis (1999)	Analyzing organizational project alternatives for agile manufacturing processes: an analytical network approach	International Journal of Production Research	284	14.2
6	Brandenburg et al. (2014)	Quantitative models for sustainable supply chain management: Developments and directions	European Journal of Operational Research	277	55.4
7	Meade and Presley (2002)	R&D project selection using the analytic network process	IEEE Transactions on Engineering Management	270	15.88
8	Jharkharia and Shankar (2007)	Selection of logistics service provider: An analytic network process (ANP) approach	Omega	267	22.25
9	Büyüközkan and Çifçi (2012)	A novel hybrid MCDM approach based on fuzzy DEMATEL, fuzzy ANP and fuzzy TOPSIS to evaluate green suppliers	Expert Systems with Applications	253	36.14
10	Yüksel and Dagdeviren (2007)	Using the analytic network process (ANP) in a SWOT analysis - A case study for a textile firm	Information Sciences	243	20.25
11	Sarkis (1998)	Evaluating environmentally conscious business practices	European Journal of Operational Research	236	11.24
12	Agarwal et al. (2006)	Modelling the metrics of lean, agile and leagile supply chain: An ANP-based approach	European Journal of Operational Research	234	18
13	Gencer and Gürpinar (2007)	Analytic network process in supplier selection: A case study in an electronic firm	Applied Mathematical Modelling	212	17.67
14	Ravi et al. (2005)	Analyzing alternatives in reverse logistics for end-of-life computers: ANP and balanced scorecard approach	Computers & Industrial Engineering	206	14.71
15	Wu (2008)	Choosing knowledge management strategies by using a combined ANP and DEMATEL approach	Expert Systems with Applications	204	18.55
16	Kuo et al. (2010)	Integration of artificial neural network and MADA methods for green supplier selection	Journal of Cleaner Production	195	21.67
17	Liou et al. (2007)	Airline safety measurement using a hybrid model	Journal of Air Transport Management	195	16.25
18	Chung et al. (2005)	Analytic network process (ANP) approach for product mix planning in semiconductor fabricator	International Journal of Production Economics	182	13
19	Shyur and Shih (2006)	A hybrid MCDM model for strategic vendor selection	Mathematical and Computer Modelling	179	13.77
20	Demirtas and Üstün (2008)	An integrated multiobjective decision making process for supplier selection and order allocation	Omega	176	16

Abbreviations: PY = publishing year; TC = total citations; TCY = total citations per year.

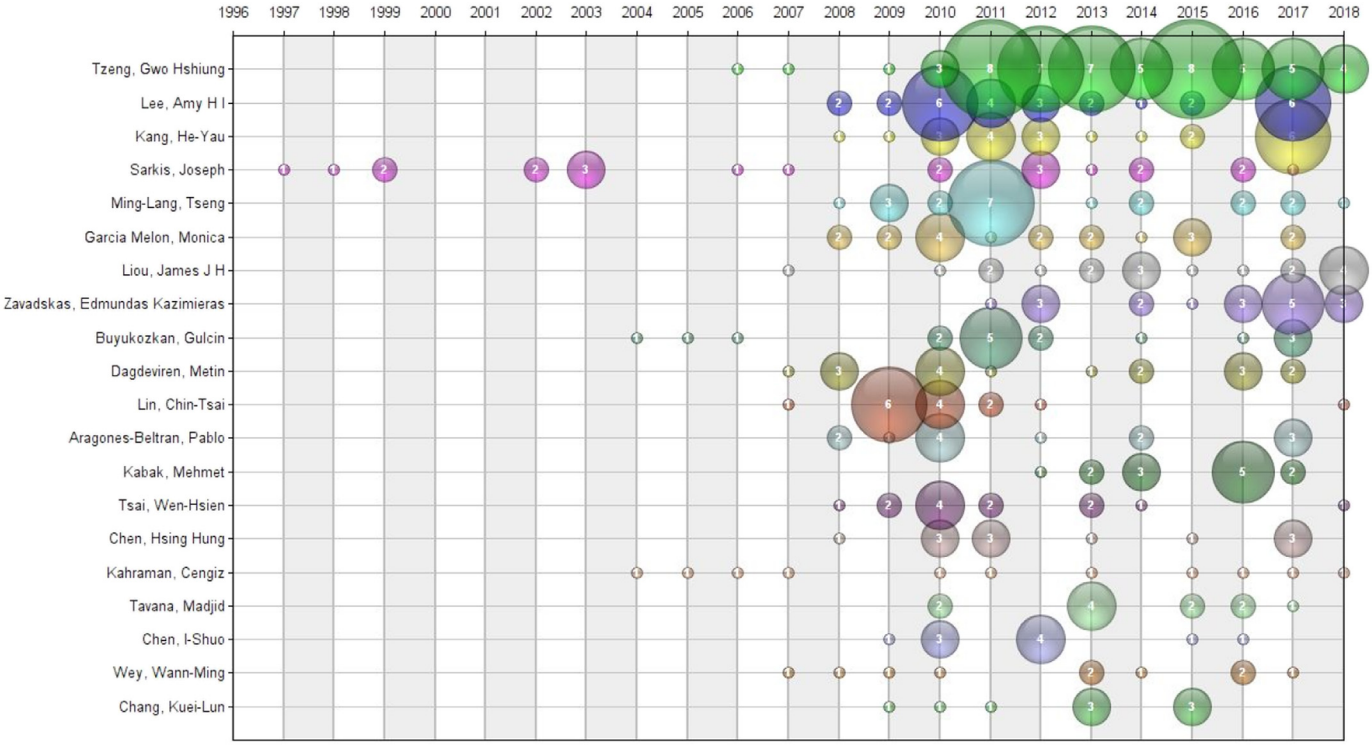

Fig. 9. Bubble chart of top 20 productive authors by year.

3.7. Analysis of most cited published documents

Among 1485 articles about the ANP published during 1997–2018, 48 (3.23%) articles were cited more than 100 times, 107 (7.21%) articles were cited between 50 and 99 times, and 189 (12.73%) of articles were cited between 25 and 49 times. Table 7 shows the top 20 most cited publications related to ANP research. The earliest paper was published in 1998, and the latest one was published in 2014. Four of these top 20 articles were published in the *European Journal of Operational Research*. The most highly cited article, entitled "A strategic decision framework for green supply chain management", was published in the *Journal of Cleaner Production* in 2003. It was authored by Sarkis (2003), who has the most articles (4 articles) on the list and harvested 513 citations. In this paper, Sarkis used the ANP to model and solve a strategic decision framework with the goal of improving the green supply chain which incorporated environmental conscious business practices. Product life cycle, operational life cycle, performance measures and environmentally influential organizational policy were considered to be the criteria.

The article entitled "A fuzzy optimization model for QFD planning process using analytic network approach" (Kahraman et al., 2006) is the second-most cited article, with 307 total citations. This study employed the fuzzy ANP to incorporate customer needs (CNs) and product technical requirements (PTRs) systematically into the product design phase in fuzzy QFD, which integrated the fuzzy logic approach, and the coefficients of the mixed integer linear programming model, which was established to optimize target improvements, were obtained from the fuzzy ANP approach. The results obtained through the fuzzy AHP, the fuzzy ANP, the crisp AHP and the crisp ANP were compared with each other and indicated that the ranking of PTRs was almost the same for the first two methods, but decision-makers could obtain the solution with the highest degree of membership by using fuzzy methods, which indicates the largest possible improvement for the PTRs.

In the third position, the article entitled "Using analytic network process and goal programming for interdependent information system project selection" by Lee and Kim (2000) has achieved 294 citations. The authors combined the ANP with a zero-one goal programming (ZOGP) model for information system project selection, and the interdependencies among information system (IS) projects had not been considered before this paper was presented. The ANP was applied prior to goal programming formulation to prioritize the criteria and objectives. The coefficients of the ZOGP model were equal to the weights obtained from the ANP model.

The fourth-most cited paper is "Product planning in quality function deployment using a combined analytic network process and goal programming approach" (Karsak et al., 2003), which was cited 286 times. It presented a ZOGP methodology that includes priorities of PTRs in the house of quality (HOQ) derived from the ANP and illustrated this decision procedure based on the pencil design example by Shillito (1994).

Meade and Sarkis (1999) introduced a decision methodology and structure for manufacturing agility improvement in the article "Analyzing organizational project alternatives for agile manufacturing processes: an analytical network approach", which was the earliest article among the top 5 most cited articles, published in 1999. The agility dimensions including "input", "output", "control" and "mechanism", and the determinants and enablers of agility were provided. A literature review of AHP and ANP was also introduced and the ANP was chosen to structure the agile configuration management decision model. The framework and methodology presented in this paper could help decision-makers filter the techniques and philosophies involved in agility according to their needs.

In addition to the total citations, it should be noted that the recently published article "Quantitative models for sustainable supply chain management: Developments and directions" authored by Brandenburg et al. (2014), has gained 55.4 citations per year since

2014. It is followed by the article "A novel hybrid MCDM approach based on the fuzzy DEMATEL, fuzzy ANP and fuzzy TOPSIS to evaluate green suppliers" (Büyüközkan and Çifçi, 2012), which was published in *Expert Systems with Applications* in 2012 with 36.14 citations per year.

3.8. Analysis of hot papers and hot topics for each year

Each year's hot papers were sorted out, and the most-cited hot papers of each year are listed in Table 8. Popular topics are presented in Table 9. According to the definition of hot paper introduced in Section 2, 145 papers were obtained. There are no articles in 2000, since none of the ANP-related articles published during 1999 and 2000 were cited in 2000. To reveal the hot topics of each year, the author keywords for these 145 publications were analyzed, and 15 main topics were ultimately determined. These 15 topics can be categorized into several typical application fields: manufacturing industry, environmental industry, power and energy industry, healthcare, logistics, e-business, research and development (R&D), finance and banking.

There are two hot papers that have been the most-cited hot papers for two consecutive years, as shown in Table 8. One of these papers is titled "Selection of logistics service provider: An analytic network process (ANP) approach" (Jharkharia and Shankar, 2007), which gained 23 citations in 2008 and is also one of the top 20 most cited articles listed in Table 7. The other paper is "Using a hybrid MCDM model to evaluate firm environmental knowledge management in uncertainty" (Tseng, 2011), which received 38 citations in 2012.

The numbers in each column of Table 9 represent the number of articles covering the topics listed in the first row. The following topics are the most popular and have gained increasing attention in the last three years: sustainability; environmental management, which includes waste management, green management and hazardous substance management; and supply chain including (green) supply chain management and supplier selection. Among the 145 hot papers, the article titled "Evaluating environmentally conscious business practices" authored by Sarkis (1998) is the first paper that utilized the ANP as a modelling approach to analyze the impact of various organizational alternatives on major environmentally conscious business practices (ECBP). This paper is also one of the top 20 most cited articles, ranking 11th, as shown in Table 7. For supplier selection, the paper "Evidential supplier selection based on DEMATEL and game theory" (Liu et al., 2018b) published in 2018

Table 8
The most cited hot papers of each year.

Year	Author, (PY)	Title	Journal	CY	TC	ACPP
1998	Meade et al. (1997)	Justifying strategic alliances and partnering: A prerequisite for virtual enterprising	Omega	1	49	2.23
1999	Sarkis (1998)	Evaluating environmentally conscious business practices	European Journal of Operational Research	2	236	11.24
2001	Lee and Kim (2000)	Using analytic network process and goal programming for interdependent information system project selection	Computers & Operations Research	1	294	15.47
2002 (1)	Partovi (2001)	An analytic model to quantify strategic service vision	International Journal of Service Industry Management	1	58	3.22
2002 (2)	Sekitani and Takahashi (2001)	A unified model and analysis for AHP and ANP	Journal of the Operations Research Society of Janpan	1	27	1.5
2003	Mikhailov and Singh (2003)	Fuzzy analytic network process and its application to the development of decision support systems	IEEE Transactions on Systems, man, and Cybernetics, Part C (Applications and Reviews)	2	102	6.38
2004	Karsak et al. (2003)	Product planning in quality function deployment using a combined analytic network process and goal programming approach	Computers & Industrial Engineering	5	286	17.88
2005 (1)	Büyüközkan et al. (2004)	Determining the importance weights for the design requirements in the house of quality using the fuzzy analytic network approach	International Journal of Intelligent Systems	4	100	6.67
2005 (2)	Yurdakul (2004)	AHP as a strategic decision-making tool to justify machine tool selection	Journal of Materials Processing Technology	4	104	6.93
2006	Ravi et al. (2005)	Analyzing alternatives in reverse logistics for end-of-life computers: ANP and balanced scorecard approach	Computers & Industrial Engineering	5	206	14.71
2007	Jharkharia and Shankar (2007)	Selection of logistics service provider: An analytic network process (ANP) approach	Omega	6	267	22.25
2008	Jharkharia and Shankar (2007)	Selection of logistics service provider: An analytic network process (ANP) approach	Omega	23	267	22.25
2009	Demirtas and Üstün (2008)	An integrated multiobjective decision making process for supplier selection and order allocation	Omega	16	176	16
2010	Tsai and Chou (2009)	Selecting management systems for sustainable development in SMEs: A novel hybrid model based on DEMATEL, ANP, and ZOGP	Expert Systems with Applications	15	149	14.9
2011	Tseng (2011)	Using a hybrid MCDM model to evaluate firm environmental knowledge management in uncertainty	Applied Soft Computing	14	94	11.75
2012	Tseng (2011)	Using a hybrid MCDM model to evaluate firm environmental knowledge management in uncertainty	Applied Soft Computing	38	94	11.75
2013	Büyüközkan and Çifçi (2012)	A novel hybrid MCDM approach based on fuzzy DEMATEL, fuzzy ANP and fuzzy TOPSIS to evaluate green suppliers	Expert Systems with Applications	15	253	36.14
2014	Yang et al. (2013)	A VIKOR technique based on DEMATEL and ANP for information security risk control assessment	Information Sciences	15	75	12.5
2015	Brandenburg et al. (2014)	Quantitative models for sustainable supply chain management: Developments and directions	European Journal of Operational Research	51	277	55.4
2016 (1)	Hashemi et al. (2015)	An integrated green supplier selection approach with analytic network process and improved Grey relational analysis	International Journal of Production Economics	26	81	20.25
2016 (2)	Kahraman et al. (2015)	Fuzzy Multicriteria Decision-Making: A Literature Review	International Journal of Computational Intelligence Systems	26	101	25.25
2017	Kou et al. (2016)	Pairwise comparison matrix in multiple criteria decision making	Technological and Economic Development of Economy	26	46	15.33
2018	Liu et al. (2018b)	Evidential Supplier Selection Based on DEMATEL and Game Theory	International Journal of Fuzzy Systems	21	22	22

Abbreviations: PY = publishing year; CY = number of citations of that year; TC = total citations; TCY = total citations per year.

Table 9
Hot topics of ANP related hot papers.

Year (No. of articles)	Topics	Strategic Decision Making	Decision Support	Health	Environmental Management	Sustainability	Power and Energy	Supply Chain Management	Quality	Location Selection	Logistics	e-business	R&D	Business Strategy	Finance and Banking	Improvement of ANP Method
1998 (1)		1														
1999 (2)				1	1											
2001 (1)			1													
2002 (2)									1							1
2003 (2)			1										1			
2004 (2)		1						1	1							
2005 (2)		1														
2006 (3)						1					1					
2007 (3)									1		1			1		
2008 (4)		1						1	1		1			1		
2009 (11)			2					2					2		1	
2010 (12)			1		3	1		4	1	2			1	1		
2011 (10)		1	2		3		1		1				1			
2012 (14)		1	2	1	5	2						1		3	1	2
2013 (12)		2		1	5	1	1	2				1				1
2014 (11)		2			4	2	1	4						2		1
2015 (15)					3	1	1	5	1		2		2			1
2016 (20)			2	1	6	4	2	7		2	1				1	
2017 (20)			2		6	2	1	10	1	2		1				2
2018 (19)			2	1	9	2	3	10	1	2						1

gained 21 citations in 2018, while its total number of citations is 22. A new evidential ANP methodology based on game theory was proposed that enables efficiently addressing supplier management under an uncertain environment and combining DEMATEL to determine the weight of criteria.

Almost every year, there are hot topics about quality, although the number of articles that mentioned quality is small. Relevant papers authored by Karsak et al. (2003), which have been introduced in Section 3.7, and the article authored by Büyüközkan et al. (2004) have received 286 and 100 total citations, respectively. These two articles use the ANP and the fuzzy ANP to prioritize design requirements in the house of quality (HoQ). Power and energy, especially renewable energy, are also inspiring sources for the ANP, which has attracted interest in recent five years. A review written by Mardani et al. (2017) and published in 2017 received 9 citations in 2017 and 15 citations in 2018, providing a review on the application and use of decision-making approaches to energy management problems. Nine of 196 published papers related to energy management employed the ANP and fuzzy ANP techniques, and the ANP approach was applied to various fields, such as strategic environmental assessment, land management, green management, climate change, construction and environmental management. It should be noted that e-business is an emerging industry that has been a hot topic in 2012, 2013 and 2017; Chen et al. (2016a) used DANP to analyze consumer decision-making in the context of e-business, and effective suggestions were given for internet vendors. This paper was cited 11 times in 2017 and 4 times in 2018.

Health has always been an essential topic of human life, healthcare, hospital service, and medical treatment. In 1999, Carter et al. (1999) compared three decision making techniques, namely, the ANP, the AHP and the Markov process in the evaluation of the optimal post-lumpectomy treatment strategy and found that these three methods can be useful tools to the clinician. Both the AHP and ANP required less development time than did the Markov process, but the Markov process can provide detailed results. After 11 years, health has once again become a hot topic.

The research trend and hotspots of the ANP can be concluded by combining the analysis of author keywords and hot papers. Fig. 6 and Table 9 present the changes of topics over time, “DEMATEL” has been used at a high frequency since 2010 and has continued to grow in the last four years. “Supply chain management”, one of the hot topics, which also includes “supplier selection”, was rarely used as an author keyword before 2009. Many hot papers published during the last ten years (2009–2018) have also emphasized this topic, especially those published in the last three years. For the topic of “quality”, “quality function deployment” did not get considerable use until 2010, and it has always attracted scholars’ attention as a hot topic despite its low-frequency use as a keyword. It is shown that “GIS” and “sustainability” were frequently used after 2010. “GIS” was usually applied to the location selection, which has been an area of focus in the past three years, and “sustainability” was always mentioned frequently in hot papers according to the data provided in Table 9. “Environment management” and “power and energy” have been discussed as hot topics since 2010. As humans and society become more concerned about topics such as the environment, energy and sustainable development, which are closely related to human life and the earth, research may place more emphasis on these areas.

4. Implications

More than 90% of the publications related to the analytic network process were published during the period from 2009 to 2018, and the number of studies has continued to grow since 2009. The analytic network process is attracting increasing research

interest and effort. Combining the analysis results explained in Section 3, this section provides some implications for the development status and future applications of the ANP from the perspectives of research areas, collaboration, and future application trends.

4.1. A research areas perspective

According to the data provided in Table 2, operations research and management science are the dominant research areas where the analytic network process is applied. Operations research and management science both pursue improved decision-making and efficiency, and the analytic network process is one of the effective methods that helps make multi-criteria decisions. Therefore, the ANP can be explored more deeply to develop extension methods or integrated methods, such as the fuzzy ANP and the generalized ANP (Zhu et al., 2015) for various application scenarios, such as scheduling (Li and Wang, 2017; Yazgan et al., 2010; Mirabedini and Iranmanesh, 2014), supply chain management (Sarkis, 2003; Hung, 2011), and facility location (Kuo, 2011; Guneri et al., 2009).

Collaborations among different research fields offer more space for analytic network process research. In addition to focusing on the strong cross-relationships among operations research, computer science, environmental science and engineering, weak cross-relationships among business, economics, energy and fuel, mathematics and other research categories should also be considered. Strong relationships indicate general study directions, while weak relationships imply potential research perspectives.

4.2. A cooperation perspective

The analysis results show that Taiwan, mainland China, Turkey, Iran and the United States are the countries/regions that contributed the highest number of publications related to the ANP. The United States has the most collaborative relationships with other countries while Serbia, Japan, Philippines and Singapore have relatively few publications and less cooperation with other countries. National Chiao Tung University is the most productive institution and has established a large collaborative network with various institutions. The collaboration matrix map (Table 7) and the DDA cluster map (Table 8) indicate the potential cooperation opportunities between different countries and institutions. Scholars from different countries or institutions can work together to promote the research progress in the ANP research field.

4.3. A future application trends perspective

The hot spots of ANP research vary over time. The integrated results presented in Section 3.3, Section 3.7, and Section 3.8 indicate that quality is an enduring hot application while energy and e-business are the emerging hot topics that have attracted scholars' interests in recent years. Sustainability, environmental management and the supply chain including green supply chain management and supplier selection, are the most popular hot topics and have gained increasing attention in the last three years.

The control of quality plays a key role in manufacturing, which is one of the reasons why quality can always be concering. Quality management will continue to be a hot topic in the future as long as manufacturing still exists. With the concept of cleaner production, new energy, sustainable development, and environmental protection being constantly raised and discussed, the ANP began to be applied to strategic environment assessment (Kabak et al., 2014), green supply chain management (Chatterjee et al., 2018; Malviya and Kant, 2016), the hazardous substance management (Hsu and Hu, 2009). Since topics including sustainability, environmental management and green supply chain management just have been frequently discussed in recent years, future research topics will apply the analytic network process to evaluate the decisions.

5. Conclusions and limitations

The characteristics of ANP research from 1996 to 2018 based on the WoS database were analyzed and hot topics were identified by using bibliometric techniques in this paper. A total of 1485 articles and reviews were evaluated based on the following factors: global contributions, productivity of countries/regions, research areas, major journals, prolific authors, representative papers with high number of citations, and major institutions that contributed to the published literature related to ANP research. Author keywords were also analyzed to show the expanding and shifting focus in ANP research.

The growth rate of the publications related to the ANP has been relatively stable in the first thirteen years (1996–2008) and a steep rise occurred in 2009, with the high rate of increase of 81.58%. More than 90% of the literature was published between 2009 and 2018. This finding reflects the fact that the study of the ANP has received wide attention since 2009.

In the aspect of research areas, "operations research and management Science" dominates the research area list with 373 articles, followed by "computer science, artificial intelligence", "engineering, industrial", "environmental sciences".

In the reference to the journals, *Expert Systems with Applications* was the most productive journal publishing articles in ANP research (118 articles). The *Journal of Cleaner Production* ranks second among the top 20 journals, and its number of publications has increased notably since 2015. The most highly cited paper, entitled "A strategic decision framework for green supply chain management", was authored by Sarkis, J and was published in this journal. *Sustainability* takes the third place. It seems that supplier selection and supply chain management are major topics of articles that were published by these top 3 journals, and the top 3 most highly cited articles (each of these articles has gained more than 170 citations) published in the *Journal of Cleaner Production* are all related to supplier selection or supply chain management.

According to the analysis of author keywords, "MCDM" (266 times) is the most frequently used keyword (except "ANP", which was used more than 800 times), and its use has increased dramatically between 2011 and 2016. It is followed by "DEMATEL", "fuzzy ANP" and "AHP". The keywords like "supplier selection", "balanced scorecard", "quality function deployment", "GIS" and other MCDM methods imply the combinations of the ANP with different techniques. Various fuzzy techniques derived from "fuzzy logic" that are shown in the chart have been applied to many areas.

Taiwan is the most productive region, contributing a total of 436 (29.36%) articles, followed by the People's Republic of China (230). Correspondingly, National Chiao Tung University produced the highest number of articles, and it also has the highest *h*-index (28) among the top 20 institutions. The United States is the most active country that cooperates with other countries, especially China, Turkey, Germany and Iran. The top 20 most productive authors are from the top 12 most productive countries. Tzeng, G. H. is the most productive author in fields related to ANP research with 56 (3.77%) publications.

Collaborations among different research fields offer more space for analytic network process research. Potential research perspectives exist in weak relationships of research categories, such as business, economics, energy and mathematics. Strong relationships among operations research, computer science and engineering indicate general study directions. Scholars from different countries or institutions can work together to promote the research progress

in the ANP research field referring to the collaboration matrix map between different countries and institutions.

Quality is an enduring hot topic. Sustainability, environmental management and green supply chain management will be future research topics that apply the analytic network process in the evaluation of decisions. Energy and e-business are the emerging hot topics that have attracted scholars' interests in recent years.

This study can provide an overview of ANP research and help researchers identify research trends and potential collaborators. Bibliometric techniques were applied in this paper, which can help determine the characteristics and research trends related to the ANP quantitatively and qualitatively. However, there are still some limitations.

First, this method of analysis cannot explore the underlying reasons behind the results presented in this paper. Second, the data collected from the database and the indicators, such as *h*-index, impact factor and the number of citations, would need to be updated over time. In addition to the limitations of the method, insufficiently comprehensive data collection, including the selection of the database and the determination of the search query would also affect the results. Although the Web of Science covers a large number of publications, other databases such as Scopus and Google Scholar, may also include relevant publications. Because the search query would be insufficiently comprehensive, valuable publications would be consequently omitted. In terms of author keywords analysis, author keywords were not contained in some articles, which would contribute to the inaccuracy of the analysis results.

Further work should focus on a more comprehensive collection of data, a more accurate analysis of the characteristics of the literature and research trends, and more in-depth examination of the causes of the results of the analysis.

Acknowledgments

This work is financially supported by National Natural Science Foundation of China, China (Grant No. 71371170, 71572184 and 71401158), Natural Science Foundation of Zhejiang Province, China (Grant No. LY17E050023, LY18G010019, LY15G010009 and LS18G03012).

References

Yüksel, İ., Dagdeviren, M., 2007. Using the analytic network process (ANP) in a SWOT analysis - a case study for a textile firm. Inf. Sci. 177 (16), 3364–3382.
Agarwal, A., Shankar, R., Tiwari, M.K., 2006. Modeling the metrics of lean, agile and leagile supply chain: an ANP-based approach. Eur. J. Oper. Res. 173 (1), 211–225.
Alizadeh, M., Ngah, I., Hashim, M., Pradhan, B., Pour, A.B., 2018. A hybrid analytic network process and artificial neural network (ANP-ANN) model for urban earthquake vulnerability assessment. Rem. Sens. 10 (6), 34.
Bao, G.J., Fang, H., Chen, L.F., Wan, Y.H., Xu, F., Yang, Q.H., Zhang, L.B., 2018. Soft robotics: academic insights and perspectives through bibliometric analysis. Soft Robot. 5 (3), 229–241.
Brandenburg, M., Govindan, K., Sarkis, J., Seuring, S., 2014. Quantitative models for sustainable supply chain management: Developments and directions. Eur. J. Oper. Res. 233 (2), 299–312.
Büyüközkan, G., Berkol, Ç., 2011. Designing a sustainable supply chain using an integrated analytic network process and goal programming approach in quality function deployment. Expert Syst. Appl. 38 (11), 13731–13748.
Büyüközkan, G., Çifçi, G., 2011. A novel fuzzy multi-criteria decision framework for sustainable supplier selection with incomplete information. Comput. Ind. 62 (2), 164–174.
Büyüközkan, G., Çifçi, G., 2012. A novel hybrid MCDM approach based on fuzzy DEMATEL, fuzzy ANP and fuzzy TOPSIS to evaluate green suppliers. Expert Syst. Appl. 39 (3), 3000–3011.
Büyüközkan, G., Ertay, T., Kahraman, C., Ruan, D., 2004. Determining the importance weights for the design requirements in the house of quality using the fuzzy analytic network approach. Int. J. Intell. Syst. 19 (5), 443–461.
Carter, K.J., Ritchey, N.P., Castro, F., Caccamo, L.P., Kessler, E., Erickson, B.A., 1999. Analysis of three decision-making methods: a breast cancer patient as a model. Med. Decis. Mak. 19 (1), 49–57.
Chatterjee, K., Pamucar, D., Zavadskas, E.K., 2018. Evaluating the performance of suppliers based on using the R'AMATEL-MAIRCA method for green supply chain implementation in electronics industry. J. Clean. Prod. 184, 101–129.
Chen, C.H., Tzeng, G.H., 2011. Creating the aspired intelligent assessment systems for teaching materials. Expert Syst. Appl. 38 (10), 12168–12179.
Chen, H.H., Kang, H.Y., Xing, X.Q., Lee, A.H.I., Tong, Y.H., 2008. Developing new products with knowledge management methods and process development management in a network. Comput. Ind. 59 (2–3), 242–253.
Chen, H.Q., Wan, Y.H., Jiang, S.A., Cheng, Y.X., 2014. Alzheimer's disease research in the future: bibliometric analysis of cholinesterase inhibitors from 1993 to 2012. Scientometrics 98 (3), 1865–1877.
Chen, H.M., Wu, C.H., Tsai, S.B., Yu, J., Wang, J., Zheng, Y., 2016a. Exploring key factors in online shopping with a hybrid model. SpringerPlus 5 (1), 19.
Chen, H.Q., Wang, X.P., He, L., Chen, P., Wan, Y.H., Yang, L.Y., Jiang, S.A., 2016b. Chinese energy and fuels research priorities and trend: a bibliometric analysis. Renew. Sustain. Energy Rev. 58, 966–975.
Chung, S.H., Lee, A.H.I., Pearn, W.L., 2005. Analytic network process (ANP) approach for product mix planning in semiconductor fabricator. Int. J. Prod. Econ. 96 (1), 15–36.
Chung, C.C., Chao, L.C., Lou, S.J., 2016. The establishment of a green supplier selection and guidance mechanism with the ANP and IPA. Sustainability 8 (3), 24.
Coban, A., Ertis, I.F., Cavdaroglu, N.A., 2018. Municipal solid waste management via multi-criteria decision making methods: a case study in Istanbul, Turkey. J. Clean. Prod. 180, 159–167.
Dehdasht, G., Zin, R.M., Ferwati, M.S., Abdullahi, M.M., Keyvanfar, A., McCaffer, R., 2017. DEMATEL-ANP risk assessment in oil and gas construction projects. Sustainability 9 (8), 24.
Demirel, N.Ç., Yücenur, G.N., Demirel, T., Muşdal, H., 2012. Risk-based evaluation of Turkish agricultural strategies using fuzzy AHP and fuzzy ANP. Human. Ecol. Risk. Asses. 18 (3), 685–702.
Demirtas, E.A., Üstün, Ö., 2008. An integrated multiobjective decision making process for supplier selection and order allocation. Omega 36 (1), 76–90.
Ergu, D., Kou, G., Shi, Y., Shi, Y., 2014. Analytic network process in risk assessment and decision analysis. Comput. Oper. Res. 42, 58–74.
Ervural, B.C., Zaim, S., Demirel, O.F., Aydin, Z., Delen, D., 2018. An ANP and fuzzy TOPSIS-based SWOT analysis for Turkey's energy planning. Renew. Sustain. Energy Rev. 82, 1538–1550.
Faust, O., Hagiwara, Y., Hong, T.J., Lih, O.S., Acharya, U.R., 2018. Deep learning for healthcare applications based on physiological signals: a review. Comput. Methods Progr. Biomed. 161, 1–13.
Feng, I.M., Chen, J.H., Zhu, B.W., Xiong, L., 2018. Assessment of and improvement strategies for the housing of healthy elderly: improving quality of life. Sustainability 10 (3), 32.
Gencer, C., Gürpinar, D., 2007. Analytic network process in supplier selection: a case study in an electronic firm. Appl. Math. Model. 31 (11), 2475–2486.
Gigović, L.J., Pamučar, D., Bajić, Z., Milićević, M., 2016. The combination of expert judgment and GIS-MAIRCA analysis for the selection of sites for ammunition depots. Sustainability 8 (4), 30.
Gómez-Navarro, T., Ribó-Pérez, D., 2018. Assessing the obstacles to the participation of renewable energy sources in the electricity market of Colombia. Renew. Sustain. Energy Rev. 90, 131–141.
Guneri, A.F., Cengiz, M., Seker, S., 2009. A fuzzy anp approach to shipyard location selection. Expert Syst. Appl. 36 (4), 7992–7999.
Güngör, Z., Delice, E.K., Kesen, S.E., 2011. New product design using FDMS and FANP under fuzzy environment. Appl. Soft Comput. 11 (4), 3347–3356.
Hashemi, S.H., Karimi, A., Tavana, M., 2015. An integrated green supplier selection approach with analytic network process and improved Grey relational analysis. Int. J. Prod. Econ. 159, 178–191.
Ho, W., 2008. Integrated analytic hierarchy process and its applications - a literature review. Eur. J. Oper. Res. 186 (1), 211–228.
Ho, W.R.J., Tsai, C.L., Tzeng, G.H., Fang, S.K., 2011. Combined DEMATEL technique with a novel MCDM model for exploring portfolio selection based on CAPM. Expert Syst. Appl. 38 (1), 16–25.
Hsu, C.W., Hu, A.H., 2009. Applying hazardous substance management to supplier selection using analytic network process. J. Clean. Prod. 17 (2), 255–264.
Hsu, C.W., Kuo, T.C., Shyu, G.S., Chen, P.S., 2014. Low carbon supplier selection in the hotel industry. Sustainability 6 (5), 2658–2684.
Hsu, W.C.J., Tsai, M.H., Tzeng, G.H., 2018. Exploring the best strategy plan for improving the digital convergence by using a hybrid MADM model. Technol. Econ. Dev. Econ. 24 (1), 164–198.
Hu, S.K., Lu, M.T., Tzeng, G.H., 2015. Improving mobile commerce adoption using a new hybrid fuzzy MADM model. Int. J. Fuzzy Syst. 17 (3), 399–413.
Hu, S.K., Liou, J.J.H., Lu, M.T., Chuang, Y.C., Tzeng, G.H., 2018. Improving NFC technology promotion for creating the sustainable education environment by using a hybrid modified MADM model. Sustainability 10 (5), 24.
Hung, S.J., 2011. Activity-based divergent supply chain planning for competitive advantage in the risky global environment: a DEMATEL-ANP fuzzy goal programming approach. Expert Syst. Appl. 38 (8), 9053–9062.
Jharkharia, S., Shankar, R., 2007. Selection of logistics service provider: an analytic network process (ANP) approach. Omega 35 (3), 274–289.
Kabak, M., Burmaoğlu, S., Kazançoğlu, Y., 2012. A fuzzy hybrid MCDM approach for professional selection. Expert Syst. Appl. 39 (3), 3516–3525.
Kabak, M., Kose, E., Kirilmaz, O., Burmaoglu, S., 2014. A fuzzy multi-criteria decision making approach to assess building energy performance. Energy Build. 72, 382–389.

Kahraman, C., Ertay, T., Büyüközkan, G., 2006. A fuzzy optimization model for QFD planning process using analytic network approach. Eur. J. Oper. Res. 171 (2), 390–411.
Kahraman, C., Onar, S.C., Oztaysi, B., 2015. Fuzzy multicriteria decision-making: a literature review. Int. J. Comput. Intell. Syst. 8 (4), 637–666.
Kang, H.Y., Hung, M.C., Pearn, W.L., Lee, A.H.I., Kang, M.S., 2011. An integrated multi-criteria decision making model for evaluating wind farm performance. Energies 4 (11), 2002–2026.
Kang, H.Y., Lee, A.H.I., Yang, C.Y., 2012. A fuzzy ANP model for supplier selection as applied to IC packaging. J. Intell. Manuf. 23 (5), 1477–1488.
Kannan, D., 2018. Role of multiple stakeholders and the critical success factor theory for the sustainable supplier selection process. Int. J. Prod. Econ. 195, 391–418.
Karsak, E.E., Sozer, S., Alptekin, S.E., 2003. Product planning in quality function deployment using a combined analytic network process and goal programming approach. Comput. Ind. Eng. 44 (1), 171–190.
Kou, G., Ergu, D., 2016. AHP/ANP theory and its application in technological and economic development: the 90th anniversary of Thomas L. Saaty. Technol. Econ. Dev. Econ. 22 (5), 649–650.
Kou, G., Ergu, D., Lin, C.S., Chen, Y., 2016. Pairwise comparison matrix in multiple criteria decision making. Technol. Econ. Dev. Econ. 22 (5), 738–765.
Kuo, M.S., 2011. Optimal location selection for an international distribution center by using a new hybrid method. Expert Syst. Appl. 38 (6), 7208–7221.
Kuo, R.J., Wang, Y.C., Tien, F.C., 2010. Integration of artificial neural network and MADA methods for green supplier selection. J. Clean. Prod. 18 (12), 1161–1170.
Kuo, R.J., Wu, Y.H., Hsu, T.S., Chen, L.K., 2011. Improving outpatient services for elderly patients in Taiwan: a qualitative study. Arch. Gerontol. Geriatr. 53 (2), 209–217.
Kuo, T.C., Hsu, C.W., Li, J.Y., 2015. Developing a green supplier selection model by using the DANP with VIKOR. Sustainability 7 (2), 1661–1689.
Lam, J.S.L., 2015. Designing a sustainable maritime supply chain: a hybrid QFD-ANP approach. Transport. Res. E Logist. Transport. Rev. 78, 70–81.
Lam, J.S.L., Dai, J., 2015. Environmental sustainability of logistics service provider: an ANP-QFD approach. Int. J. Logist. Manag. 26 (2), 313–333.
Lee, J.W., Kim, S.H., 2000. Using analytic network process and goal programming for interdependent information system project selection. Comput. Oper. Res. 27 (4), 367–382.
Lee, A.H.I., Lin, C.Y., 2011. An integrated fuzzy QFD framework for new product development. Flex. Serv. Manuf. J. 23 (1), 26–47.
Lee, A.H.I., Kang, H.Y., Yang, C.Y., Lin, C.Y., 2010. An evaluation framework for product planning using FANP, QFD and multi-choice goal programming. Int. J. Prod. Res. 48 (13), 3977–3997.
Lee, A.H.I., Hung, M.C., Kang, H.Y., Pearn, W.L., 2012. A wind turbine evaluation model under a multi-criteria decision making environment. Energy Convers. Manag. 64, 289–300.
Lee, A.H.I., Kang, H.Y., Liou, Y.J., 2017. A hybrid multiple-criteria decision-making approach for photovoltaic solar plant location selection. Sustainability 9 (2), 21.
Li, K.L., Wang, J., 2017. Multi-objective optimization for cloud task scheduling based on the ANP model. Chin. J. Electron. 26 (5), 889–898.
Lin, F., Huang, Y.W., 2015. Factors affecting passenger choice of low cost carriers: an analytic network process approach. Tour. Manage. Perspectives. 16, 1–10.
Lin, Y.T., Lin, C.L., Yu, H.C., Tzeng, G.H., 2010. A novel hybrid MCDM approach for outsourcing vendor selection: a case study for a semiconductor company in Taiwan. Expert Syst. Appl. 37 (7), 4796–4804.
Lin, C.T., Hung, K.P., Hu, S.H., 2018. A decision-making model for evaluating and selecting suppliers for the sustainable operation and development of enterprises in the aerospace industry. Sustainability 10 (3), 21.
Liou, J.J.H., Chuang, Y.T., 2010. Developing a hybrid multi-criteria model for selection of outsourcing providers. Expert Syst. Appl. 37 (5), 3755–3761.
Liou, J.J.H., Tzeng, G.H., Chang, H.C., 2007. Airline safety measurement using a hybrid model. J. Air Transp. Manag. 13 (4), 243–249.
Liu, H.T., Tsai, Y.I., 2012. A fuzzy risk assessment approach for occupational hazards in the construction industry. Saf. Sci. 50 (4), 1067–1078.
Liu, K.M., Lin, S.H., Hsieh, J.C., Tzeng, G.H., 2018a. Improving the food waste composting facilities site selection for sustainable development using a hybrid modified MADM model. Waste Manag. 75, 44–59.
Liu, T.Y., Deng, Y., Chan, F., 2018b. Evidential supplier selection based on DEMATEL and game theory. Int. J. Fuzzy Syst. 20 (4), 1321–1333.
Lu, M.T., Hu, S.K., Huang, L.H., Tzeng, G.H., 2015. Evaluating the implementation of business-to-business m-commerce by SMEs based on a new hybrid MADM model. Manag. Decis. 53 (2), 290–317.
Malviya, R.K., Kant, R., 2016. Hybrid decision making approach to predict and measure the success possibility of green supply chain management implementation. J. Clean. Prod. 135, 387–409.
Mao, G., Zou, H., Chen, G., Du, H., Zuo, J., 2015. Past, current and future of biomass energy research: a bibliometric analysis. Renew. Sustain. Energy Rev. 52, 1823–1833.
Mardani, A., Zavadskas, E.K., Khalifah, Z., Zakuan, N., Jusoh, A., Nor, K.M., Khoshnoudi, M., 2017. A review of multi-criteria decision-making applications to solve energy management problems: two decades from 1995 to 2015. Renew. Sustain. Energy Rev. 71, 216–256.
Marzi, G., Dabić, M., Daim, T., Garces, E., 2017. Product and process innovation in manufacturing firms: a 30-year bibliometric analysis. Scientometrics 113 (2), 673–704.
Meade, L., 1998. Strategic analysis of logistics and supply chain management systems using the analytical network process. Transport. Res. E Logist. Transport. Rev. 34 (3), 201–215.
Meade, L.A., Presley, A., 2002. R&D project selection using the analytic network process. IEEE Trans. Eng. Manag. 49 (1), 59–66.
Meade, L.M., Sarkis, J., 1999. Analyzing organizational project alternatives for agile manufacturing processes: an analytical network approach. Int. J. Prod. Res. 37 (2), 241–261.
Meade, L.M., Liles, D.H., Sarkis, J., 1997. Justifying strategic alliances and partnering: a prerequisite for virtual enterprising. Omega 25 (1), 29–42.
Meho, L.I., Yang, K., 2007. Impact of data sources on citation counts and rankings of LIS faculty: Web of science versus scopus and google scholar. J. Am. Soc. Inf. Sci. Technol. 58 (13), 2105–2125.
Merigó, J.M., Yang, J.B., 2017. A bibliometric analysis of operations research and management science. Omega 73, 37–48.
Mikhailov, L., Singh, M.G., 2003. Fuzzy analytic network process and its application to the development of decision support systems. IEEE Trans. Syst. Man Cybern. C Appl. Rev. 33 (1), 33–41.
Mirabedini, S.N., Iranmanesh, H., 2014. A scheduling model for serial jobs on parallel machines with different preventive maintenance (PM). Int. J. Adv. Manuf. Technol. 70 (9–12), 1579–1589.
Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Ann. Intern. Med. 151 (4), 264–269.
Niemira, M.P., Saaty, T.L., 2004. An analytic network process model for financial-crisis forecasting. Int. J. Forecast. 20 (4), 573–587.
Nilashi, M., Ahmadi, H., Ahani, A., Ravangard, R., bin Ibrahim, O., 2016. Determining the importance of hospital information system adoption factors using fuzzy analytic network process (ANP). Technol. Forecast. Soc. Change 111, 244–264.
Pal, D.K., Ravi, B., Bhargava, L.S., 2007. Rapid tooling route selection for metal casting using QFD-ANP methodology. Int. J. Comput. Integr. Manuf. 20 (4), 338–354.
Partovi, F.Y., 2001. An analytic model to quantify strategic service vision. Int. J. Serv. Ind. Manag. 12 (5), 476–499.
Partovi, F.Y., 2006. An analytic model for locating facilities strategically. Omega 34 (1), 41–55.
Partovi, F.Y., Corredoira, R.A., 2002. Quality function deployment for the good of soccer. Eur. J. Oper. Res. 137 (3), 642–656.
Pei, Z., 2015. Intuitionistic fuzzy variables: concepts and applications in decision making. Expert Syst. Appl. 42 (22), 9033–9045.
Pei, Z., Zheng, L., 2012. A novel approach to multi-attribute decision making based on intuitionistic fuzzy sets. Expert Syst. Appl. 39 (3), 2560–2566.
Pritchard, A., 1969. Statistical bibliography or bibliometrics? J. Doc. 25 (4), 348–349.
Quezada, L.E., López-Ospina, H.A., Palominos, P.I., Oddershede, A.M., 2018. Identifying causal relationships in strategy maps using ANP and DEMATEL. Comput. Ind. Eng. 118, 170–179.
Rajesh, R., Ravi, V., 2015. Supplier selection in resilient supply chains: a grey relational analysis approach. J. Clean. Prod. 86, 343–359.
Ravi, V., Shankar, R., Tiwari, M.K., 2005. Analyzing alternatives in reverse logistics for end-of-life computers: ANP and balanced scorecard approach. Comput. Ind. Eng. 48 (2), 327–356.
Saaty, T.L., 1980. The Analytic Hierarchy Process. McGraw-Hill, New York.
Saaty, T.L., 1996. Decision Making with Dependence and Feedback: the Analytic Network Process. RWS Publications, Pittsburgh.
Saaty, T.L., 2004a. Decision making the analytic hierarchy and network processes (AHP/ANP). J. Syst. Sci. Syst. Eng. 13 (1), 1–35.
Saaty, T.L., 2004b. Fundamentals of the analytic network process dependence and feedback in decision-making with a single network. J. Syst. Sci. Syst. Eng. 13 (1), 129–157.
Saaty, T.L., 2007. Time dependent decision-making; dynamic priorities in the AHP/ANP: generalizing from points to functions and from real to complex variables. Math. Comput. Model. 46 (7–8), 860–891.
Saaty, T.L., 2008. Relative measurement and its generalization in decision making why pairwise comparisons are central in mathematics for the measurement of intangible factors the analytic hierarchy/network process (to the memory of my beloved friend professor sixto rios garcia). Revista De La Real Academia De Ciencias Exactas Fisicas Y Naturales Serie a-Matematicas 102 (2), 251–318.
Saaty, T.L., 2013. Decision Making with the Analytic Network Process: Economic, Political, Social and Technological Applications with Benefits, Opportunities, Costs and Risks, second ed. Springer, New York.
Saaty, T.L., Ergu, D., 2015. When is a decision-making method trustworthy? Criteria for evaluating multi-criteria decision-making methods. Int. J. Inf. Technol. Decis. Mak. 14 (6), 1171–1187.
Saaty, T.L., Vargas, L.G., 2005. The possibility of group welfare functions. Int. J. Inf. Technol. Decis. Mak. 4 (2), 167–176.
Sarkis, J., 1998. Evaluating environmentally conscious business practices. Eur. J. Oper. Res. 107 (1), 159–174.
Sarkis, J., 2003. A strategic decision framework for green supply chain management. J. Clean. Prod. 11 (4), 397–409.
Sekitani, K., Takahashi, I., 2001. A unified model and analysis for AHP and ANP. J. Oper. Res. Soc. Jpn. 44 (1), 67–89.
Shillito, M.L., 1994. Advanced QFD – Linking Technology to Market and Company Needs. Wiley, New York.
Shyur, H.J., 2006. COTS evaluation using modified TOPSIS and ANP. Appl. Math. Comput. 177 (1), 251–259.
Shyur, H.J., Shih, H.S., 2006. A hybrid MCDM model for strategic vendor selection. Math. Comput. Model. 44 (7–8), 749–761.

Sivasamy, K., Arumugam, C., Devadasan, S.R., Murugesh, R., Thilak, V.M.M., 2016. Advanced models of quality function deployment: a literature review. Qual. Quantity 50 (3), 1399–1414.
Tadić, S., Zečević, S., Krstić, M., 2014. A novel hybrid MCDM model based on fuzzy DEMATEL, fuzzy ANP and fuzzy VIKOR for city logistics concept selection. Expert Syst. Appl. 41 (18), 8112–8128.
Tosarkani, B.M., Amin, S.H., 2018. A possibilistic solution to configure a battery closed-loop supply chain: multi-objective approach. Expert Syst. Appl. 92, 12–26.
Tsai, W.H., Chou, W.C., 2009. Selecting management systems for sustainable development in smes: a novel hybrid model based on DEMATEL, ANP, and ZOGP. Expert Syst. Appl. 36 (2), 1444–1458.
Tsai, W.H., Leu, J.D., Liu, J.Y., Lin, S.J., Shaw, M.J., 2010. A MCDM approach for sourcing strategy mix decision in IT projects. Expert Syst. Appl. 37 (5), 3870–3886.
Tseng, M.L., 2011. Using a hybrid MCDM model to evaluate firm environmental knowledge management in uncertainty. Appl. Soft Comput. 11 (1), 1340–1352.
Tseng, M.L., Lin, R.J., Lin, Y.H., Chen, R.H., Tan, K., 2014. Close-loop or open hierarchical structures in green supply chain management under uncertainty. Expert Syst. Appl. 41 (7), 3250–3260.
van Raan, A.F.J., 2005. For your citations only? Hot topics in bibliometric analysis. Measurement: Interdisciplinary. Res. Perspective. 3 (1), 50–62.
Wallin, J.A., 2005. Bibliometric methods: pitfalls and possibilities. Basic Clin. Pharmacol. Toxicol. 97 (5), 261–275.
Wang, Y.L., Tzeng, G.H., 2012. Brand marketing for creating brand value based on a MCDM model combining DEMATEL with ANP and VIKOR methods. Expert Syst. Appl. 39 (5), 5600–5615.
Wang, Y., Xiang, C., Zhao, P., Mao, G., Du, H., 2016. A bibliometric analysis for the research on river water quality assessment and simulation during 2000-2014. Scientometrics 108 (3), 1333–1346.
Wong, J., Li, H., Lai, J., 2008. Evaluating the system intelligence of the intelligent building systems - part 1: development of key intelligent indicators and conceptual analytical framework. Autom. ConStruct. 17 (3), 284–302.
Wu, W.W., 2008. Choosing knowledge management strategies by using a combined ANP and DEMATEL approach. Expert Syst. Appl. 35 (3), 828–835.
Wu, C., Barnes, D., 2016. An integrated model for green partner selection and supply chain construction. J. Clean. Prod. 112, 2114–2132.
Wu, Y.Q., Wan, Y.H., Zhang, F.Z., 2018. Characteristics and trends of C-H activation research: a review of literature. Curr. Org. Synth. 15 (6), 781–792.
Yang, Y.P.O., Shieh, H.M., Tzeng, G.H., 2013. A VIKOR technique based on DEMATEL and ANP for information security risk control assessment. Inf. Sci. 232, 482–500.
Yang, C.L., Yuan, B.J.C., Huang, C.Y., 2015. Key determinant derivations for information technology disaster recovery site selection by the multi-criterion decision making method. Sustainability 7 (5), 6149–6188.
Yazgan, H.R., Boran, S., Goztepe, K., 2010. Selection of dispatching rules in FMS: ANP model based on BOCR with choquet integral. Int. J. Adv. Manuf. Technol. 49 (5–8), 785–801.
Yeung, A.W.K., Mocan, A., Atanasov, A.G., 2018. Let food be thy medicine and medicine be thy food: a bibliometric analysis of the most cited papers focusing on nutraceuticals and functional foods. Food Chem. 269, 455–465.
Yurdakul, M., 2004. AHP as a strategic decision-making tool to justify machine tool selection. J. Mater. Process. Technol. 146 (3), 365–376.
Zadeh, L.A., 1965. Fuzzy sets. Inf. Control 8 (1), 338–353.
Zaim, S., Sevkli, M., Camgöz-Akdağ, H., Demirel, O.F., Yayla, A.Y., Delen, D., 2014. Use of ANP weighted crisp and fuzzy QFD for product development. Expert Syst. Appl. 41 (9), 4464–4474.
Zhang, Z.H., Awasthi, A., 2014. Modelling customer and technical requirements for sustainable supply chain planning. Int. J. Prod. Res. 52 (17), 5131–5154.
Zhang, D., Fu, H.Z., Ho, Y.S., 2017. Characteristics and trends on global environmental monitoring research: a bibliometric analysis based on science citation index expanded. Environ. Sci. Pollut. Control Ser. 24 (33), 26079–26091.
Zhu, B., Xu, Z.S., Zhang, R., Hong, M., 2015. Generalized analytic network process. Eur. J. Oper. Res. 244 (1), 277–288.
Zyoud, S.H., Fuchs-Hanusch, D., 2017. A bibliometric-based survey on AHP and TOPSIS techniques. Expert Syst. Appl. 78, 158–181.